

A proof of the Apéry Hankel growth conjecture

OEIS A228143, linearly shifted determinants,
and a non-D-finite generating series

with Part II: sharp Szegő asymptotics, the ratio constant K ,
and monotone Hankel envelopes, and Part III: proportional shifts

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Abstract

For the Apéry numbers $A_k = \sum_{j=0}^k \binom{k}{j}^2 \binom{k+j}{j}^2$, let $D_n = \det(A_{i+j})_{0 \leq i, j \leq n}$. The OEIS entry A228143 records a conjecture, added by Vaclav Kotesovec on 13 September 2026, that $D_n^{1/n^2} \rightarrow (1 + \sqrt{2})^4/4$. Starting from Edgar's known positive moment-density theorem, we prove this limit and the stronger estimate $\log D_n = n(n+1) \log((1 + \sqrt{2})^4/4) + O(n)$. The proof uses a polynomial lower bound for the density, monic orthogonal polynomials, and elementary Legendre–Chebyshev estimates. We also obtain an explicit growth law for $\det(A_{m(i+j)+r_n})$ when m is fixed and r_n/n converges, including nonzero linear shifts. A further consequence is that the formal ordinary generating series of D_n is not D-finite. The package includes exact determinants through $n = 200$, independent arithmetic checks, and runnable Python and C++/GMP code. The density representation is credited, not reproved; the subsequent determinant arguments are given in full.

Part II, added on 1 October 2026 (batch 73O1), answers the question Part I leaves open: the normalized norms $h_n/\Lambda^{2n} = D_n/(D_{n-1}\Lambda^{2n})$ converge, to $K = (\pi C/2) \exp(\log \varphi)$, an arcsine average of Edgar's density ($K \approx 0.633427435172$, a numerical evaluation). Hence $D_n/D_{n-1} \sim K\Lambda^{2n}$ and $\log D_n = n(n+1) \log \Lambda + (n+1) \log K + o(n)$. The proof lifts the density to the unit circle and uses a proved Szegő minimum principle; it recovers Part I's main theorem only as a corollary. Part II also gives interleaved monotone Hankel envelopes $E_m \downarrow K$ computable in $\mathbb{Q}(\sqrt{2})$, sharp constants for fixed polynomial modifications and fixed strides and shifts, a two-term inverse-index law, and a conditional transfer theorem whose hypothesis is not proved for the Apéry density. The full multiplicative prefactor of D_n remains open.

Part III (2 October 2026, batch 77P5) proves, for $H_N(r) = \det(A_{r+i+j})_{0 \leq i, j < N}$ and $r/N = s$ in compact subsets of $(3\sqrt{2}/4 - 1, \infty)$, $H_N(r) \sim M(s)N^{-1/24} e^{N^2 A(s) + NB(s)}$ with explicit A, B, M and a first correction; A is Part I's linear-shift law at stride one. Strides $m \geq 2$, smaller s and D_n ($s = 0$) are not covered.

Main result

$$\lim_{n \rightarrow \infty} D_n^{1/n^2} = \frac{17 + 12\sqrt{2}}{4} = 8.4926406871192851464 \dots$$

More precisely, some $\kappa > 0$ satisfies, for every $n \geq 0$,

$$\kappa^n \left(\frac{17 + 12\sqrt{2}}{4} \right)^{n(n+1)} \leq D_n \leq 4^n \left(\frac{17 + 12\sqrt{2}}{4} \right)^{n(n+1)}.$$

Status and attribution. This is a proof of the precise assertion still labelled a conjecture in the accessed OEIS revision, not a claim that the result has no antecedent anywhere in the literature. The essential pre-existing analytic input is G. A. Edgar's work. This report has not been peer reviewed or formalized in a proof assistant. All three Parts are AI-assisted research notes; none is peer reviewed or formalized.

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Part I. A proof of the Apéry Hankel growth conjecture

1 The selected problem and the stronger result

The Apéry numbers, OEIS A005259, are

$$A_k = \sum_{j=0}^k \binom{k}{j}^2 \binom{k+j}{j}^2, \quad A_0 = 1, \quad A_1 = 5, \quad A_2 = 73. \quad (1.1)$$

Define the leading Hankel determinants

$$D_n = \det(A_{i+j})_{0 \leq i, j \leq n}. \quad (1.2)$$

Thus the matrix has order $n + 1$, not n . The first values are 1, 48, 161856, 39002646528, ...; they agree with OEIS A228143 [2, 1].

The target is the asymptotic formula in the %F field of revision #40 of A228143, dated 13 September 2026. It is attributed to Vaclav Kotesovec and asserts

$$\lim_{n \rightarrow \infty} D_n^{1/n^2} = \frac{(1 + \sqrt{2})^4}{4}. \quad (1.3)$$

It is distinct from the divisibility and root-integrality assertions also recorded in that entry. The source status here refers to the version accessed on 18 September 2026 [1].

Use the constants

$$C = 17 + 12\sqrt{2} = (1 + \sqrt{2})^4, \quad c = C^{-1} = 17 - 12\sqrt{2}, \quad \Lambda = \frac{C}{4}. \quad (1.4)$$

The small positive number c is an interior singular point of the moment density; it is *not* the left endpoint of its support.

Theorem 1.1. *There exists $\kappa \in (0, 1]$ such that*

$$\kappa^n \Lambda^{n(n+1)} \leq D_n \leq 4^n \Lambda^{n(n+1)} \quad (n \geq 0). \quad (1.5)$$

Consequently,

$$\log D_n = n(n+1) \log \Lambda + O(n), \quad (1.6)$$

$$D_n^{1/n^2} = \Lambda \exp(O(1/n)), \quad (1.7)$$

$$\left(\frac{D_n}{D_{n-1}} \right)^{1/(2n)} \rightarrow \Lambda \quad (n \rightarrow \infty). \quad (1.8)$$

In particular, (1.3) is true.

The proof will bound each monic orthogonal-polynomial norm separately. This is stronger than trying to estimate one large determinant after replacing its entries by asymptotic approximations. The latter operation is generally unsafe; Section 6 gives an exact counterexample to that method.

2 The known moment density and a global lower bound

Known input. Edgar’s Theorem 1, Proposition 4, Notation 6, Propositions 25–26, and Corollary 27 [3] give an integrable density φ on $[0, C]$ such that

$$A_k = \int_0^C x^k \varphi(x) dx, \quad \int_0^C \varphi(x) dx = 1. \quad (2.1)$$

It is positive and real analytic on $(0, c)$ and (c, C) , with

$$\varphi(x) = -\frac{6}{\pi^2} \log x + o(1) \quad (x \downarrow 0), \quad (2.2)$$

$$\varphi(x) = \frac{(C-x)^{1/2}}{2^{5/4} C \pi^2} + O((C-x)^{3/2}) \quad (x \uparrow C). \quad (2.3)$$

At c , on each side, it is a nonzero combination of convergent Frobenius solutions with leading orders $0, \frac{1}{2}, 1$. Here our C, c correspond to Edgar’s c, c_0 .

In particular, for positive distance t from c on either side, the first nonzero term gives

$$\varphi(c \pm t) = a_{\pm} t^{\alpha_{\pm}} (1 + o(1)), \quad a_{\pm} > 0, \quad \alpha_{\pm} \in \{0, \frac{1}{2}, 1\}. \quad (2.4)$$

Indeed, the three basis solutions start with distinct powers. A nonzero solution has a first nonzero coefficient, and positivity fixes the sign of its leading term. No value of $\varphi(c)$ needs to be assigned, and equality of its one-sided limits is not required below.

Lemma 2.1 (Polynomial minorant). *Let $Q(x) = (x-c)(x-C)$. There is $\eta > 0$ such that*

$$\varphi(x) \geq \eta Q(x)^2 \quad \text{for almost every } x \in [0, C]. \quad (2.5)$$

Proof. Consider $R(x) = \varphi(x)/Q(x)^2$ away from c, C . On each of $(0, c)$ and (c, C) this is continuous and positive. Since $cC = 1$, $Q(0) = 1$, and (2.2) implies $R(x) \rightarrow +\infty$ as $x \downarrow 0$. Equation (2.3) gives $R(x) \asymp (C-x)^{-3/2} \rightarrow +\infty$ as $x \uparrow C$. At c , (2.4) gives $R(c \pm t) \asymp t^{\alpha_{\pm}-2} \rightarrow +\infty$. Choose small neighborhoods of these three points on which R is bounded below by a positive constant. On the remaining compact pieces, its positive continuous minimum is also positive. The least of these bounds is an admissible η . $\square$

The interior singularity is the delicate point in this argument. Positivity alone would not justify a strictly positive uniform lower bound for the density. Squaring a polynomial that vanishes at c and C avoids that incorrect inference. The constant η is established to exist; this report does not claim a certified numerical value for it.

[Added 5 October 2026, batch 97G: a contrast with another Apéry-like sequence. For the level-seven sequence A279619, with terms a_n , Section 17 of the collection report a279619-level-seven-gamma-constant (in generating-functions-and-asymptotics/oeis-sequence-asymptotics/) proves that the Hankel matrix $(a_{i+j})_{i,j \geq 0}$ is strictly totally positive of order five but not totally nonnegative of order six (its Theorem 17.1), and that no tail $(a_{N+n})_{n \geq 0}$ is a Stieltjes moment sequence, because the generating function is singular at $z = -1$ (its Theorem 17.17 and Proposition 17.16); a positive Hamburger measure, if one exists, is supported in $[-1, 27]$ (its Proposition 17.19). By (2.1) the Apéry numbers are a Stieltjes moment sequence, so no such obstruction arises here. No theorem is shared: that report’s fixed-order Hankel asymptotics (its Theorems 17.3 and 17.12) keep the order r fixed as the shift $n \rightarrow \infty$, whereas Part III lets the order N and the shift $r = sN$ grow together.]

3 Elementary norm estimates and the proof

3.1 A moment determinant is a product of minimum norms

For a nonzero real polynomial p , positivity of φ on an interval gives $\int p(x)^2 \varphi(x) dx > 0$. Thus its polynomial inner product is positive definite. Let p_k be its monic orthogonal polynomial of degree k , and put

$$h_k = \int_0^C p_k(x)^2 \varphi(x) dx = \min_{\substack{p \in \mathbb{R}[x] \text{ monic} \\ \deg p = k}} \int_0^C p(x)^2 \varphi(x) dx. \quad (3.1)$$

The minimum follows by writing $p = p_k + q$, $\deg q < k$, and using orthogonality. The change of basis from $1, x, \dots, x^n$ to $p_0, \dots, p_n$ is unitriangular. Its determinant is one, so taking Gram determinants gives

$$D_n = \prod_{k=0}^n h_k, \quad h_0 = 1. \quad (3.2)$$

These are the usual monic-norm identities; compare DLMF §18.2 [7]. The argument above also proves $D_n > 0$ for every n .

3.2 The Lebesgue lower comparison

Lemma 3.1. *For $L > 0$ and $k \geq 0$, the minimum squared norm of a monic polynomial of degree k for Lebesgue measure on $[0, L]$ is*

$$\ell_k(0; L) = \frac{L^{2k+1}}{(2k+1) \binom{2k}{k}^2} \geq L \left(\frac{L}{4} \right)^{2k}. \quad (3.3)$$

Proof. On $[0, 1]$, the polynomial

$$R_k(t) = \frac{1}{k!} \frac{d^k}{dt^k} [t^k(t-1)^k]$$

is orthogonal to all lower-degree polynomials by k integrations by parts. Its leading coefficient is $\binom{2k}{k}$. Another integration by parts gives

$$\int_0^1 R_k(t) t^k dt = \int_0^1 t^k (1-t)^k dt = \frac{(k!)^2}{(2k+1)!}.$$

Orthogonality therefore implies $\int_0^1 R_k(t)^2 dt = 1/(2k+1)$. Dividing by its leading coefficient and scaling $x = Lt$ proves the equality in (3.3) and the minimizing property.

For the inequality, set $u_k = (2k+1) \binom{2k}{k}^2 / 16^k$. Direct cancellation gives $u_0 = 1$ and

$$\frac{u_{k+1}}{u_k} = 1 - \frac{1}{4(k+1)^2} < 1.$$

Thus $(2k+1) \binom{2k}{k}^2 \leq 16^k$, as needed. □

If p is monic of degree k , then pQ is monic of degree $k+2$. Lemmas 2.1 and 3.1 give

$$\int_0^C p(x)^2 \varphi(x) dx \geq \eta \int_0^C (p(x)Q(x))^2 dx \geq \eta \ell_{k+2}(0; C) \geq \eta C \Lambda^{2k+4}.$$

Consequently, with

$$\kappa = \eta C \Lambda^4 > 0, \quad h_k \geq \kappa \Lambda^{2k} \quad (k \geq 0). \quad (3.4)$$

The case $k = 0$, together with $h_0 = 1$, shows $\kappa \leq 1$.

3.3 The Chebyshev upper comparison

The Chebyshev polynomial T_k satisfies $T_k(\cos \theta) = \cos(k\theta)$. The recurrence $T_{k+1}(t) = 2tT_k(t) - T_{k-1}(t)$ shows that its leading coefficient is 2^{k-1} for $k \geq 1$. Hence

$$q_k(x) = 2\Lambda^k T_k\left(\frac{2x}{C} - 1\right)$$

is monic of degree k and has absolute value at most $2\Lambda^k$ on $[0, C]$. Since the measure has total mass one, the minimum in (3.1) satisfies

$$h_k \leq \int_0^C q_k(x)^2 \varphi(x) dx \leq 4\Lambda^{2k} \quad (k \geq 1). \quad (3.5)$$

Proof of Theorem 1.1. Multiply (3.4) and (3.5) over $k = 1, \dots, n$ and use (3.2) with $h_0 = 1$. Since $2 \sum_{k=1}^n k = n(n+1)$, this is (1.5). Taking logarithms gives (1.6); dividing by n^2 gives (1.7). Finally $D_n/D_{n-1} = h_n$, and the individual bounds imply

$$\Lambda \kappa^{1/(2n)} \leq h_n^{1/(2n)} \leq \Lambda 4^{1/(2n)},$$

which proves (1.8). □

Remark 3.2. The proof does not assert that h_n/Λ^{2n} converges. It proves this sequence is bounded above and bounded away from zero. Nor does (1.6) give a full multiplicative asymptotic equivalent for D_n . [Added 1 October 2026, batch 73O1: Part II proves that h_n/Λ^{2n} converges, to the constant $K \approx 0.6334$ of (10.2) (Theorem 10.1), so $D_n/D_{n-1} \sim K\Lambda^{2n}$ and $\log D_n = n(n+1)\log \Lambda + (n+1)\log K + o(n)$. A full multiplicative equivalent for D_n is still not established: Part II determines the prefactor only conditionally (Theorem 18.1).]

4 An explicit law for linearly shifted determinants

The same comparison idea gives more than fixed shifts. For integers $m \geq 1$ and $r \geq 0$, define

$$H_n^{(m,r)} = \det(A_{m(i+j)+r})_{0 \leq i,j \leq n}. \quad (4.1)$$

Here m is the stride in the moment indices. The shift r is held constant throughout each matrix, but may vary from one matrix size to another.

Define, for $\sigma \geq 0$,

$$\Psi(\sigma) = \frac{1}{2} [(\sigma+2)^2 \log(\sigma+2) - 2(\sigma+1)^2 \log(\sigma+1) + \sigma^2 \log \sigma], \quad (4.2)$$

with $0^2 \log 0 := 0$.

Theorem 4.1 (Linear-shift growth law). *Fix a positive integer m . If r_n are nonnegative integers and $r_n/n \rightarrow \rho \in [0, \infty)$, then*

$$\lim_{n \rightarrow \infty} \frac{\log H_n^{(m,r_n)}}{n^2} = (m+\rho) \log C - \Psi(\rho/m). \quad (4.3)$$

Equivalently,

$$\lim_{n \rightarrow \infty} (H_n^{(m,r_n)})^{1/n^2} = C^{m+\rho} \exp(-\Psi(\rho/m)).$$

In particular, every sublinear shift $r_n = o(n)$ has limit $C^m/4$.

For example,

$$\lim_{n \rightarrow \infty} \det(A_{2i+2j})_{0 \leq i, j \leq n}^{1/n^2} = \frac{C^2}{4}, \quad (4.4)$$

$$\lim_{n \rightarrow \infty} \det(A_{i+j+n})_{0 \leq i, j \leq n}^{1/n^2} = \frac{16C^2}{81\sqrt{3}}. \quad (4.5)$$

The constant $C^m/4$ for a stride m is not $(C/4)^m$. The proof below also provides a finite-size comparison with a completely explicit determinant. [Added 2 October 2026, batch 77P5: for stride $m = 1$ and ρ in compact subsets of (s_*, ∞) , $s_* = 3\sqrt{2}/4 - 1$, Part III refines this limit to a full asymptotic equivalent with a first correction (Corollary 22.2 and Theorem 28.1); its quadratic coefficient is the limit (4.3) (Remark 21.1). Part III indexes by the matrix order $N = n + 1$.]

4.1 The power-weight model

For $s > -1$ and $L > 0$, put

$$\mathcal{J}_n(s; L) = \det \left(\frac{L^{i+j+s+1}}{i+j+s+1} \right)_{0 \leq i, j \leq n}. \quad (4.6)$$

This is the moment determinant of $y^s dy$ on $[0, L]$. Its monic minimum norms are

$$\ell_k(s; L) = \frac{L^{2k+s+1}}{(2k+s+1) \binom{2k+s}{k}^2}, \quad \binom{2k+s}{k} = \prod_{j=1}^k \frac{k+s+j}{j}. \quad (4.7)$$

To verify (4.7), replace the Rodrigues polynomial in Lemma 3.1 by

$$R_{k,s}(t) = \frac{t^{-s}}{k!} \frac{d^k}{dt^k} [t^{k+s}(t-1)^k].$$

It is a polynomial of degree k , with leading coefficient $\binom{2k+s}{k}$. Integration by parts proves orthogonality for weight t^s . The boundary terms vanish because $s > -1$. Its pairing with t^k is the beta integral $\int_0^1 t^{k+s}(1-t)^k dt$; dividing by the leading coefficient and scaling the interval gives (4.7).

We will use both its norm product and its Cauchy determinant form:

$$\mathcal{J}_n(s; L) = \prod_{k=0}^n \ell_k(s; L) = L^{(n+1)(n+s+1)} \frac{\prod_{j=0}^n (j!)^2}{\prod_{i=0}^n \prod_{j=0}^n (s+1+i+j)}. \quad (4.8)$$

For completeness, the second equality follows by factoring powers of L and applying the Cauchy identity

$$\det \left(\frac{1}{x_i + y_j} \right) = \frac{\prod_{i < j} (x_j - x_i)(y_j - y_i)}{\prod_{i, j} (x_i + y_j)}$$

with $x_i = i$, $y_j = s + 1 + j$. One elementary proof of that identity clears the common denominator: the resulting polynomial is alternating in the x_i and in the y_j , so is divisible by both Vandermonde products. Total degrees agree. The remaining constant is one, by induction on the residue at $x_n = -y_n$, starting with the 1×1 case.

4.2 Uniform comparison with the power-weight model

Proposition 4.2. *For fixed $m \geq 1$ and $T < \infty$, uniformly for integers $0 \leq r \leq Tn$,*

$$\log H_n^{(m,r)} = \log \mathcal{J}_n(r/m; C^m) + O_{m,T}(n) \quad (n \geq 1). \quad (4.9)$$

Proof. Set $L = C^m$, $a = c^m$, $s = r/m$, and change variables $y = x^m$ in (2.1). The moments in (4.1) are those of $y^s w_m(y) dy$, where

$$w_m(y) = \frac{1}{m} y^{1/m-1} \varphi(y^{1/m}), \quad 0 < y < L. \quad (4.10)$$

The map is smooth with nonzero derivative near a and L . Thus the one-sided finite orders at a and the square-root order at L are unchanged. At zero,

$$w_m(y) \sim \frac{6}{m^2 \pi^2} y^{1/m-1} \log(1/y) \rightarrow +\infty.$$

The argument of Lemma 2.1 therefore supplies $\eta_m > 0$ with $w_m(y) \geq \eta_m[(y-a)(y-L)]^2$ almost everywhere.

Set $\beta = -1 + 1/(2m) \in (-1, 0)$. Near zero, $w_m(y)/y^\beta \rightarrow 0$, and this ratio is bounded on the remaining intervals, including their one-sided limits at a and L . Hence a constant $K_m > 0$ also exists such that

$$\eta_m[(y-a)(y-L)]^2 \leq w_m(y) \leq K_m y^\beta. \quad (4.11)$$

Neither constant depends on r or n .

Let $h_k(s)$ denote the monic minimum norm for $y^s w_m(y) dy$. Multiplication by the monic quadratic on the left of (4.11), and a minimizing trial polynomial for the upper power weight, give

$$\eta_m \ell_{k+2}(s; L) \leq h_k(s) \leq K_m \ell_k(s + \beta; L).$$

The parameter $s + \beta$ is greater than -1 . Multiplying over $k = 0, \dots, n$ yields

$$\eta_m^{n+1} \mathcal{J}_n(s; L) \frac{\ell_{n+1}(s; L) \ell_{n+2}(s; L)}{\ell_0(s; L) \ell_1(s; L)} \leq H_n^{(m,r)} \leq K_m^{n+1} \mathcal{J}_n(s + \beta; L). \quad (4.12)$$

We check that both corrections to $\log \mathcal{J}_n(s; L)$ are $O(n)$. Write $b = -\beta \in (0, 1)$. Formula (4.8) gives

$$\log \frac{\mathcal{J}_n(s + \beta; L)}{\mathcal{J}_n(s; L)} = (n+1)\beta \log L - \sum_{i,j=0}^n \log \left(1 - \frac{b}{s+1+i+j} \right).$$

For $t \geq 1$, $0 \leq -\log(1 - b/t) \leq b/((1-b)t)$. Counting pairs with fixed $i + j$ gives

$$\sum_{i,j=0}^n \frac{1}{s+1+i+j} \leq \sum_{i,j=0}^n \frac{1}{1+i+j} \leq 2n+1.$$

Thus the logarithmic upper correction is $O_m(n)$, uniformly in $s \geq 0$.

For the lower correction, $s \leq (T/m)n$. At $k = 0, 1, n+1, n+2$, formula (4.7) shows $|\log \ell_k(s; L)| = O_{m,T}(n)$. For $k = 0, 1$ this is immediate. For the other two indices use

$$1 \leq \binom{2k+s}{k} \leq \frac{(2k+s)^k}{k!} \leq \left(\frac{e(2k+s)}{k} \right)^k,$$

whose logarithm is $O_{m,T}(n)$ since $k \geq n+1$ and $s = O(n)$. Here $k! \geq (k/e)^k$ follows by integrating $\log x$ on $[1, k]$. The remaining factors in (4.7) have logarithms $O(n)$ as well. Applying these estimates to (4.12) proves (4.9). $\square$

4.3 Evaluating the limiting constant

Lemma 4.3. *If $s_n \geq 0$ and $s_n/n \rightarrow \sigma \geq 0$, then*

$$\frac{1}{n^2} \log \mathcal{J}_n(s_n; L) \rightarrow (1 + \sigma) \log L - \Psi(\sigma). \quad (4.13)$$

Proof. Taking logarithms in (4.8), the factorial part satisfies

$$2 \sum_{j=0}^n \log(j!) = n^2 \log n - \frac{3}{2} n^2 + O(n \log(n+1)). \quad (4.14)$$

For example, interchange the sums to obtain $2 \sum_{k=1}^n (n+1-k) \log k$ and compare the sums of $\log k$ and $k \log k$ with their integrals. For the denominator, Riemann sums give

$$\frac{1}{n^2} \sum_{i,j=0}^n \log(s_n + 1 + i + j) - \log n \rightarrow \int_0^1 \int_0^1 \log(\sigma + x + y) dx dy. \quad (4.15)$$

When $\sigma > 0$ the integrand is continuous. When $\sigma = 0$, split off $i + j \leq \delta n$. The normalized absolute contribution there is bounded by $O(\delta^2(1 + |\log \delta|)) + o(1)$: count the $i + j = k$ pairs and use $\log(n/(k+1))$ as an upper bound for the absolute logarithm for large n . Outside this corner, convergence is uniform. Letting $\delta \downarrow 0$ justifies (4.15) also at $\sigma = 0$.

Two integrations, with $t^2 \log t$ assigned its continuous value at zero, yield

$$\int_0^1 \int_0^1 \log(\sigma + x + y) dx dy = \Psi(\sigma) - \frac{3}{2}.$$

The two $\frac{3}{2}$ terms cancel between this expression and (4.14). The power of L in (4.8) contributes $(1 + \sigma) \log L$, proving the claim. $\square$

Proof of Theorem 4.1. The convergence $r_n/n \rightarrow \rho$ implies $r_n \leq Tn$ eventually for some finite T . Apply Proposition 4.2, then Lemma 4.3 with $s_n = r_n/m$, $L = C^m$, and $\sigma = \rho/m$. The error $O(n)$ disappears after division by n^2 . Finally $\Psi(0) = \log 4$. $\square$

For fixed m, r , (4.9) and the norm formula also give $\log H_n^{(m,r)} = n(n+1) \log(C^m/4) + O_{m,r}(n)$. To see this without a further asymptotic theorem, changing the power parameter from 0 to a fixed $s = r/m$ changes $\log \mathcal{J}_n$ by $O_s(n)$, using $\log(1 + s/t) \leq s/t$ and the same reciprocal sum estimate as above. Lemma 3.1 and the Chebyshev upper comparison for Lebesgue measure then bound $\log \mathcal{J}_n(0; L) - n(n+1) \log(L/4)$ by $O_L(n)$. The stride m is fixed throughout these statements; no uniform assertion for $m \rightarrow \infty$ is made.

5 A generating-function consequence

Theorem 5.1. *The sequence (D_n) is not P-recursive over $\mathbb{C}$. Its formal ordinary generating series $F(z) = \sum_{n \geq 0} D_n z^n$ is not D-finite. Moreover, for every fixed real s , the series $\sum_{n \geq 0} D_n z^n / (n!)^s$ has radius of convergence zero.*

Proof. Suppose that, for all sufficiently large integers n ,

$$\sum_{j=0}^q P_j(n) D_{n+j} = 0, \quad P_j \in \mathbb{C}[n], \quad P_q \neq 0. \quad (5.1)$$

The case $q = 0$ contradicts $D_n > 0$. For $q \geq 1$, the leading coefficient polynomial has no zeros at sufficiently large integers. Ratios of the other coefficient polynomials to it have at most polynomial growth. Therefore some $K > 0$, integer $d \geq 0$, and N satisfy

$$|D_{n+q}| \leq K n^d \max_{0 \leq j < q} |D_{n+j}| \quad (n \geq N).$$

Taking running maxima and increasing K if necessary gives $|D_n| \leq K_0 K_1^n (n!)^d$. Thus $\log D_n = O(n \log n)$, contradicting (1.6), since $\Lambda > 1$. This proves the first assertion.

A polynomial-coefficient linear differential equation for F would, by coefficient comparison, give a nontrivial polynomial-coefficient recurrence for its coefficients for all sufficiently large indices. Hence F cannot be D-finite. This argument is about formal power series and does not assume that F converges.

Finally,

$$\frac{1}{n} \log \frac{D_n}{(n!)^s} = n \log \Lambda + O(\log(n+1)) \rightarrow +\infty.$$

The root test gives radius zero for every fixed s . □

Divergence alone would *not* have proved non-D-finiteness: a formal D-finite series can have radius zero. The polynomial-recurrence growth bound is the essential extra argument.

6 Why entrywise asymptotics do not determine this limit

Consider, for $0 \leq a < C$, the elementary moments

$$m_k^{(a)} = \int_a^C x^k dx = \frac{C^{k+1} - a^{k+1}}{k+1}.$$

For each fixed $0 < a < C$,

$$\frac{m_k^{(a)}}{m_k^{(0)}} = 1 - (a/C)^{k+1}.$$

Thus these two moment sequences agree to every algebraic relative asymptotic order in $1/k$, with an exponentially small relative difference. Nevertheless their determinant limits differ. Translation by a is a unitriangular change of the monomial basis, so it does not change a Gram determinant. Applying Lemma 3.1 on the translated interval of length $C - a$ gives the exact formula

$$\det(m_{i+j}^{(a)})_{0 \leq i, j \leq n} = \frac{(C-a)^{(n+1)^2}}{\prod_{k=0}^n (2k+1) \binom{2k}{k}^2}. \quad (6.1)$$

Its n^2 -root limit is $(C-a)/4$, rather than $C/4$. Indeed, the same lower and upper monic-norm comparisons used above prove this limit directly.

This is a counterexample to a tempting inference about Hankel determinants, not to Kotesovec's conjecture. It explains why a single-entry asymptotic formula cannot safely be substituted into a growing determinant. In the Apéry case, incorrectly discarding the short interval $(0, c)$ would likewise suggest the wrong constant

$$\frac{C-c}{4} = 6\sqrt{2} = 8.485281374\dots, \quad \text{instead of} \quad \frac{C}{4} = 8.492640687\dots$$

The full positive support and the global comparison in Lemma 2.1 retain this small but consequential interval.

7 Exact computations and independent checks

The calculations supplement, but are not used as premises of, the proof. Moments were generated using the classical recurrence

$$(k+1)^3 A_{k+1} = (2k+1)(17k^2 + 17k + 5)A_k - k^3 A_{k-1}, \quad A_0 = 1, A_1 = 5, \quad (7.1)$$

recorded in Edgar’s equation (1) and OEIS A005259 [3, 2]. A symmetric fraction-free elimination pass on the 201×201 Hankel matrix produced $D_0, \dots, D_{200}$ as exact integers. At pivot k , the update was

$$a_{ij}^{\text{new}} = \frac{a_{kk}a_{ij} - a_{ik}a_{kj}}{p}, \quad i, j > k, \quad (7.2)$$

where p is the preceding pivot, initially 1. Every division was checked for zero remainder. Positivity of all leading minors means no pivoting is needed in this application. Successive pivots give the successive leading determinants. There were exactly $200 \cdot 201 \cdot 202/6 = 1,353,400$ checked divisions in the symmetric update; D_{200} has 37,308 decimal digits. The recorded GMP determinant phase took about 124 seconds in this runtime, a measurement rather than a performance guarantee.

Check	Count	Result
Recurrence vs. defining binomial sum ($0 \leq k \leq 400$)	401	Pass
Displayed OEIS prefix ($D_0, \dots, D_9$)	10	Pass
Python vs. GMP ($D_0, \dots, D_{40}$)	41	Pass
Independent rational Gaussian determinants ($n \leq 12$)	13	Pass
Independent modular determinants (11 sizes, 3 primes)	33	Pass
Shifted Desnanot–Jacobi identities	64	Pass
Power-weight Cauchy determinants, rational parameters	32	Pass
Central-binomial inequality ($0 \leq k \leq 500$)	501	Pass

Table 1: Exact verification results. The full integer divisions were also checked during generation.

Table 2 and Figure 1 use logarithms of these exact integers, not floating-point determinants. Hankel determinant formulas are known to be ill-conditioned in direct numerical use; DLMF §18.2(ix) explicitly warns about this issue [7]. The displayed decimal values are numerical evaluations, not interval-certified bounds. The exact data and exact-arithmetic checks are provided separately.

n	D_n^{1/n^2}	$(D_n/D_{n-1})^{1/(2n)}$
1	48.0000000000	6.9282032303
2	20.0577493937	7.6202968816
5	11.9387651922	8.1187083186
10	10.0601045949	8.3009844581
20	9.2400493095	8.3954447392
50	8.7832622606	8.4538328708
100	8.6366837149	8.4732336106
200	8.5643598268	8.4829390793

Table 2: Root diagnostics computed from exact integers; the common limit is $\Lambda = 8.492640687119 \dots$

The shift experiment file records exact samples at $n = 10, 20, 40, 80$. The three families have $(m, r_n) = (1, n), (2, 0)$, and $(2, n)$. For each sample the shift is fixed throughout that matrix. Samples through $n = 20$ were also computed with the Python implementation. Their numerical roots, compared to the limits in Theorem 4.1, appear in Table 3.

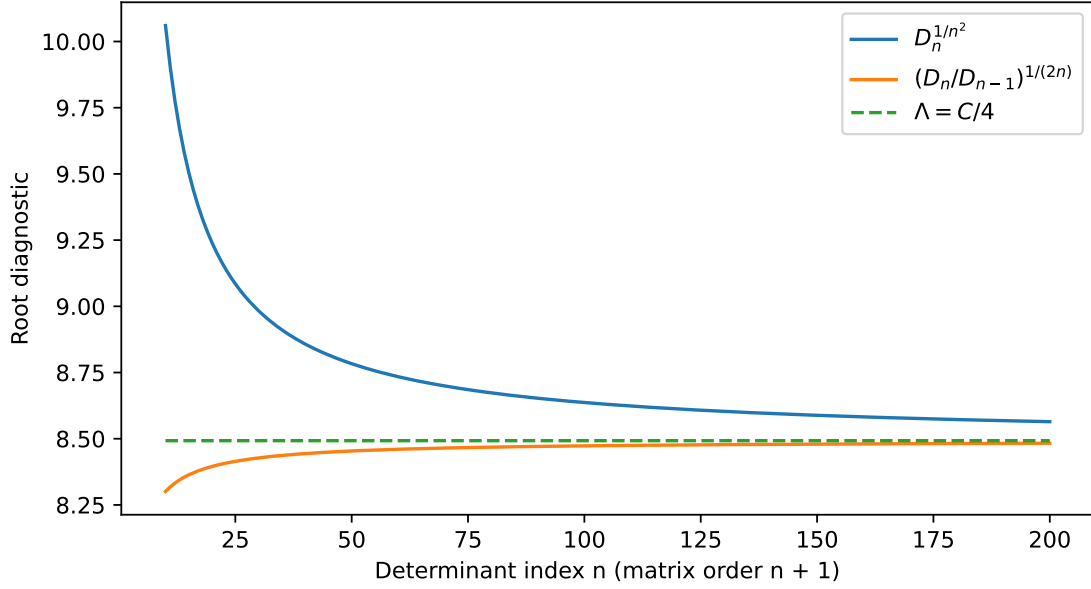


Figure 1: Two root diagnostics from exact determinants, for $10 \leq n \leq 200$. The horizontal line is the proved limit Λ . The finite-sample placement of the curves above or below it is an observation, not a claim of monotonicity or a separate theorem.

(m, ρ)	Proved limit	$n = 20$	$n = 40$	$n = 80$
(1, 1)	131.6072514	154.1754400	142.5207807	136.9744192
(2, 0)	288.4997834	360.3947981	322.3570941	304.9315450
(2, 1)	6075.4685812	8406.8961687	7151.9818542	6593.0434765

Table 3: Numerical values of $(H_n^{(m, r_n)})^{1/n^2}$ for $r_n = \rho n$. The limits follow from the theorem, not from extrapolation.

8 What has and has not been established

The established conclusion is a proof of the stated OEIS growth limit, using a previously known positive density theorem, with a stronger $O(n)$ error in the logarithm. The same method yields the explicit linear-shift law and the non-D-finiteness corollary. In particular, the work is not only a search for a numerical counterexample: the target conjecture turns out to admit a positive proof.

The analysis does not evaluate the best lower-envelope constant η , determine a limit for h_n/Λ^{2n} , or give a full multiplicative asymptotic equivalent such as $D_n \sim K\Lambda^{n^2}B^n n^\gamma$. [Added 1 October 2026, batch 73O1: Part II determines the limit of h_n/Λ^{2n} and the linear term of $\log D_n$ (Theorem 10.1); in the notation of this sentence it gives $B = \Lambda K$ with K of (10.2), up to a factor $e^{o(n)}$. The multiplicative constant, the exponent γ (and a possible power of $\log n$) remain open, and η is not evaluated there either; see Remark 10.4 and Section 20.] None of the other arithmetic assertions on A228143 is being claimed here as a newly solved problem. The literature check establishes the status of the specific accessed entry, not worldwide priority for every consequence of Edgar's theorem.

A natural continuation is to determine the finer normalized norm and determinant asymptotics while treating the interior singular point and the logarithmic endpoint accurately. The elementary comparisons already settle the leading growth questions; finer constants require additional arguments, not extrapolation from the data. [Added 1 October 2026, batch 73O1: Part II carries out the first step of this continuation, the normalized norms and the linear term, with the logarithmic endpoint and the interior singular point handled through the Szegő condition (Lemma 11.1). The finer prefactor and the oscillation caused by the interior singular point remain open (Section 20).]

Part II. Sharp Szegő asymptotics for Apéry Hankel determinants

Ratio limits, monotone Hankel envelopes, polynomial modifications, and inverse growth

Research report prepared for the Provelt project (with ChatGPT), dated 1 October 2026; added to this report on 1 October 2026 in batch 7301 of Provelt's incoming-report intake.

Sections 1–8 and Appendix A form the original report (Part I) of 18 September 2026, unchanged except for dated notes pointing to this addition, a front matter that now announces both Parts, and the place of Appendix A, which now follows Part II together with Part II's own appendices. Part II was written against Part I as printed above (repository commit bc4d1fa2b). It answers the question Part I left open in the remark after its proof of Theorem 1.1 and in its Section 8: the normalized norms h_n/Λ^{2n} converge, and their limit is an explicit integral constant K . Section 9.2 records the provenance and fixes the notation of the addition.

Summary (the manuscript's abstract). Let $A_k = \sum_{j=0}^k \binom{k}{j}^2 \binom{k+j}{j}^2$ and $D_n = \det(A_{i+j})_{0 \leq i,j \leq n}$, the sequence OEIS A228143. Part I proves its quadratic logarithmic growth. Part II identifies the next, linear term and establishes a genuine equivalent for D_n/D_{n-1} , with an explicit integral constant K . In the normalization used in the OEIS ratio plot, the limit is $\Lambda K \approx 5.379471608281$, where $\Lambda = (17 + 12\sqrt{2})/4$. It constructs an interleaved, nonincreasing sequence of finite Hankel expressions converging from above to K , and expresses the discrepancy of the original oscillating ratios by square-summable reflection coefficients. Further results give sharp equivalents for every fixed nonnegative polynomial modification, fixed strides and shifts, and a two-term inverse-index law. An independently proved conditional transfer theorem explains how finer reflection-coefficient expansions produce powers of n , powers of $\log n$, and a lattice correction in inverse asymptotics. The analytic input is Edgar's positive moment density; the Szegő minimum principle and all subsequent reductions used here are proved explicitly. Finite exact checks and reproducible high-precision computations accompany the Part.

The principal refinement. Put $C = 17 + 12\sqrt{2}$ and $\Lambda = C/4$. For Edgar's density φ on $[0, C]$, define

$$K = \frac{\pi C}{2} \exp \left\{ \frac{1}{\pi} \int_0^C \frac{\log \varphi(x)}{\sqrt{x(C-x)}} dx \right\}.$$

Then $0 < K < \infty$ and

$$\frac{D_n}{D_{n-1}} \sim K \Lambda^{2n}, \quad \log D_n = n(n+1) \log \Lambda + (n+1) \log K + o(n).$$

The integral gives numerically $K \approx 0.633427435172$. The decimal is a numerical evaluation, not an interval-certified enclosure.

Scope and attribution. This is a sharper continuation of the result of Part I, not a new proof claimed for the already-established leading growth limit. The general Szegő theory is classical. Absolute bibliographic priority for these Apéry specializations is not established. The proofs are conventional mathematical proofs, not proof-assistant formalizations.

9 What Part I proved, and what changes

Part II uses the Apéry numbers A_k and the determinants D_n of Part I, (1.1) and (1.2), with the additional convention $D_{-1} = 1$. The matrix in D_n has order $n + 1$, and the first determinants are 1, 48, 161856, 39002646528, ..., in agreement with OEIS A005259 and A228143 [2, 1].

The OEIS entry accessed on 1 October 2026 still labels

$$D_n^{1/n^2} \longrightarrow \frac{(1 + \sqrt{2})^4}{4}$$

as a conjecture. However, Part I (*A proof of the Apéry Hankel growth conjecture*, 18 September 2026) already proves the stronger bound (1.6), $\log D_n = n(n + 1) \log \Lambda + O(n)$ with $\Lambda = (17 + 12\sqrt{2})/4$, through the two-sided estimate (1.5) of Theorem 1.1. It also handles linearly growing shifts at the leading n^2 scale (Theorem 4.1). Re-proving (1.6) would not constitute progress over Part I.

The present target is the sharper question suggested by the ratio plot linked from A228143 [4]: does a normalized *consecutive ratio* converge, and what is its limit? This question is distinct from a separately worded conjecture in an OEIS formula field. The plot visibly has oscillations; a numerical plateau alone does not identify the limit reliably. In Part I's notation it is the question whether h_n/Λ^{2n} converges, which Part I explicitly left open.

Quantity	Part I	Part II
$\log D_n$	Quadratic term, error $O(n)$	Exact linear coefficient, error $o(n)$
D_n/D_{n-1}	Its $2n$ -th root converges	A genuine equivalent with constant K
Ratio fluctuations	Numerical oscillations	Exact monotone-envelope decomposition
Fixed modified moments	Leading growth in selected families	Sharp norm constants for polynomial factors, strides and shifts
Inverse growth	Not the target of Part I	Constant correction to the inverse index

9.1 Source audit and limits on novelty claims

The comparison uses repository commit bc4d1fa2b849b644461e3b8c597d77aa1c54b9fe (the manuscript called it a “tree”; it is a commit) and the Apéry growth article printed above as Part I, whose source at that commit is the text of 18 September 2026. The OEIS entry, its linked plot, Edgar's moment-density paper [3], and the classical unit-circle orthogonal-polynomial literature [5] were checked on 1 October 2026. Searches for Apéry Hankel ratio asymptotics and an Apéry Szegő constant did not reveal this sharper formula in the material examined. This is a bounded literature check, not proof that no prior publication contains an equivalent result.

The importance of the refinement is mathematical rather than rhetorical: an unknown $O(n)$ term allows an unknown exponential factor in D_n . The constant K identifies that factor, determines a normalized generating function's radius, and fixes a constant shift in the inverse index. Part II does not claim a new general Szegő theorem or a field-changing breakthrough.

9.2 Provenance and conventions of Part II

Part II is built from one manuscript, number 41 of batch 73O1: *Sharp Szegő asymptotics for Apéry Hankel determinants* (1 October 2026), delivered as `apery_hanke1_research.zip` in arrival commit `bdf1a1a73` (an archive that shares its name, but not its content, with the archive that delivered Part I) and placed in commit `9df4ba51a`; its pin is the commit `bc4d1fa2b` named above. Its programs, data and figures are shipped with the prefix `02-szego-`; the README of this report lists them. The merge made the following choices.

- The manuscript’s restatement of Edgar’s density input (its §3, the local laws at 0, c and C) and of the definitions of A_k and D_n is not repeated: Part II cites Part I’s Section 2 instead, and the displays (2.2), (2.3), (2.4), (1.1), (1.2) and (1.6) of Part I stand for the manuscript’s restated copies (whose labels are therefore not carried over). Every theorem, proposition, lemma, corollary, remark, proof, example, table, figure and question of the manuscript is printed.
- The manuscript’s citation of Part I as an external repository article is replaced by references to Part I’s numbered statements; its two OEIS references are merged with Part I’s bibliography entries, and its Edgar reference with Part I’s.
- Part I’s Theorem 1.1 is recovered by the Szegő route only as a corollary (Remark 10.4); it is not claimed as new.
- File names in the text are the shipped names; the delivered programs themselves still read and write their delivery names (see the README).

Notation. Part II keeps the manuscript’s symbols except for the two renamings marked $\star$ in Table 4, made because Part I already uses those letters for different objects. Where a letter is shared, the table prints the Part I meaning beside the Part II one.

Table 4: Symbols shared by the two Parts.

Symbol	Part I	Part II
C, c, Λ	(1.4)	the same
$h_k, H_n^{(m,r)}, w_m$	(3.1), (4.1), (4.10)	the same objects ($h_n = h_n[1]$; r fixed in Section 16)
K	a generic constant in the proof of Theorem 5.1; a placeholder in the shape $K\Lambda^{n^2}B^n n^\gamma$ of Section 8	the Szegő ratio constant (10.2), $K \approx 0.6334$
$\star \mathcal{K}_m$	(K_m): the upper constant $w_m \leq K_m y^\beta$ in (4.11)	the stride constant (16.2); the manuscript writes K_m
$\star \xi$	(η): the minorant constant of Lemma 2.1	the log log exponent (18.2); the manuscript writes η
L	a free interval length in Lemma 3.1 and (4.6)	$L = \log \Lambda$, (10.1)
B, B_m	a placeholder base in Section 8	$B = L + \log K$ (17.1); separately $B_m = C^m$ (16.2) (B_m is <i>not</i> B at index m)
Ψ	$\Psi(\sigma)$ of (4.2)	$\Psi_C(z)$, the exterior conformal map (15.5)
α	$\alpha_\pm \in \{0, \frac{1}{2}, 1\}$, Frobenius exponents in (2.4)	α_j , reflection coefficients (12.2); the exponents are called $\beta_\pm$
F	the generating series $\sum D_n z^n$ (Theorem 5.1)	the polygonal interpolation of $\log D_n$ (Section 17); ${}_2F_1(\frac{1}{3}, \frac{2}{3}; 1; z)$ in Appendix B
κ, η	constants of (3.4) and Lemma 2.1	not used

The bracket $\langle f \rangle_C$ of (10.2) is an arcsine average, not an average against $\varphi(x) dx$; C_0 in Proposition 18.2 is a constant term, not C .

10 The sharp asymptotic theorem

Set

$$C = 17 + 12\sqrt{2}, \quad c = C^{-1}, \quad \Lambda = C/4, \quad L = \log \Lambda. \quad (10.1)$$

Let φ be the positive moment density in Edgar's theorem, so $A_k = \int_0^C x^k \varphi(x) dx$. Write

$$\langle f \rangle_C = \frac{1}{\pi} \int_0^C \frac{f(x)}{\sqrt{x(C-x)}} dx, \quad K = \frac{\pi C}{2} \exp \langle \log \varphi \rangle_C. \quad (10.2)$$

The bracket is an average with respect to a probability measure, not an average with respect to $\varphi(x) dx$.

Theorem 10.1 (Sharp Apéry ratio and logarithmic growth). *The integral in (10.2) is absolutely convergent, and $0 < K \leq 2$. As $n \rightarrow \infty$,*

$$\frac{D_n}{D_{n-1}} = K \Lambda^{2n} (1 + o(1)), \quad (10.3)$$

$$\log D_n = n(n+1)L + (n+1) \log K + o(n), \quad (10.4)$$

$$\frac{D_{n+1}}{(1 + 1/\sqrt{2})^{8n+4} D_n} \rightarrow \Lambda K, \quad (10.5)$$

$$\frac{D_{n+1} D_{n-1}}{D_n^2} \rightarrow \Lambda^2. \quad (10.6)$$

The proof is completed in Sections 11–14. No conjecture about the shape or monotonicity of the plotted ratios is used.

Remark 10.2 (A crucial distinction). Equation (10.4) says

$$D_n = \Lambda^{n(n+1)} K^{n+1} \exp(o(n)).$$

It does *not* say $D_n \sim \Lambda^{n(n+1)} K^{n+1}$. Powers of n , powers of $\log n$, and a further multiplicative constant are all compatible with the theorem. A full all-orders expansion for D_n is not asserted here.

Corollary 10.3 (Fixed windows and quadratic normalization). *For each fixed integer $k \geq 0$,*

$$\frac{D_{n+k}}{D_n} \sim K^k \Lambda^{2kn+k(k+1)}. \quad (10.7)$$

For fixed nonnegative integers r, s ,

$$\frac{D_{n+r+s} D_n}{D_{n+r} D_{n+s}} \rightarrow \Lambda^{2rs}. \quad (10.8)$$

The series

$$B(z) = \sum_{n \geq 0} \frac{D_n}{\Lambda^{n(n+1)}} z^n \quad (10.9)$$

has radius of convergence $1/K$.

Proof. Multiply the k consecutive instances of (10.3) to obtain (10.7); the sum of their exponents is $2 \sum_{j=1}^k (n+j) = 2kn + k(k+1)$. Equation (10.8) follows by cancellation. For $b_n = D_n / \Lambda^{n(n+1)}$, (10.3) gives $b_n / b_{n-1} \rightarrow K$, hence $b_n^{1/n} \rightarrow K$ and the root test proves (10.9). $\square$

Remark 10.4 (The Szegő route recovers Theorem 1.1 of Part I). Theorem 10.1 implies Part I's Theorem 1.1 except for its explicit sandwich: (10.4) gives (1.6) and hence (1.7) and the A228143 limit (1.3), and (10.3) gives (1.8). This is a second route to a result that Part I proved first, and it is printed as a corollary, not as a new result. The two routes differ. Part I compares the density with the polynomial minorant of Lemma 2.1 and with Legendre and Chebyshev norms, which yields the two-sided estimate (1.5) for every n , with an existential κ ; Part II passes to the circle and uses the Szegő minimum principle (Theorem 12.1), which yields the constant K but no lower bound at finite n , and upper bounds only through the envelopes of Theorem 14.1. The two are consistent: (3.4) gives $h_n/\Lambda^{2n} \geq \kappa$ for every n , so every admissible κ of Part I satisfies $\kappa \leq K \approx 0.6334$, and (3.5) is compatible with $K \leq 2$.

11 The moment density satisfies the Szegő condition

Known analytic input. The input is the one stated in Part I's Section 2, from [3, Theorem 1, Proposition 4, Propositions 25–26, Corollary 27]: φ is integrable and positive on $(0, c) \cup (c, C)$, represents the moments A_k by (2.1), has the endpoint laws (2.2) at 0 and (2.3) at C , and at the interior singularity c its one-sided local expansions are nonzero combinations of convergent Frobenius solutions beginning with powers $0, 1/2, 1$. (The manuscript restated these displays; they are not repeated here.) These facts are used as input; their Heun-function positivity proof is not repeated.

Lemma 11.1. *One has*

$$\int_0^C \frac{|\log \varphi(x)|}{\sqrt{x(C-x)}} dx < \infty. \quad (11.1)$$

Proof. Near zero, (2.2) gives $\log \varphi(x) = \log \log(1/x) + O(1)$. This is integrable against $x^{-1/2} dx$, as is seen with $x = t^2$. Near C , (2.3) gives $\log \varphi(x) = \frac{1}{2} \log(C-x) + O(1)$, which is integrable against $(C-x)^{-1/2} dx$.

On either side of c , select the first nonzero coefficient among the three Frobenius branches. Their leading powers are distinct, so cancellation cannot remove that first power. Positivity gives $\varphi(c \pm t) = a_{\pm} t^{\beta_{\pm}}(1 + o(1))$ for some $a_{\pm} > 0$ and $\beta_{\pm} \in \{0, 1/2, 1\}$ (this is (2.4) of Part I, where the exponents are called $\alpha_{\pm}$). Thus the absolute logarithm is bounded by a constant times $1 + |\log t|$ there. Away from these three points the density is positive and analytic, so its logarithm is bounded on compact subintervals. Combining the bounds proves (11.1). $\square$

11.1 Hankel determinants as products of minimum norms

For a nonzero nonnegative polynomial P on $[0, C]$, let

$$D_n[P] = \det \left(\int_0^C x^{i+j} P(x) \varphi(x) dx \right)_{i,j=0}^n, \quad D_{-1}[P] = 1, \quad (11.2)$$

$$h_n[P] = \min_{\substack{p \in \mathbb{R}[x] \text{ monic} \\ \deg p = n}} \int_0^C p(x)^2 P(x) \varphi(x) dx. \quad (11.3)$$

Write $h_n = h_n[1]$. Positivity on an interval makes the polynomial inner product positive definite. Its monic orthogonal polynomial $p_n[P]$ realizes the minimum: every other monic polynomial is $p_n[P] + q$, with $\deg q < n$, and orthogonality separates the squared norms. A unitriangular change of basis in the Gram determinant therefore gives

$$D_n[P] = \prod_{j=0}^n h_j[P], \quad h_n[P] = \frac{D_n[P]}{D_{n-1}[P]}. \quad (11.4)$$

In particular, every determinant used below is positive. For $P = 1$, $h_n[1]$ is Part I's h_n of (3.1) and (11.4) is Part I's (3.2); the weighted version is the same argument.

12 A proof of the circle minimum principle

The method is classical Szegő theory; see [5] for its broader setting. The precise minimum principle needed here has a short proof. It is included to make the normalization and hypotheses explicit.

Let $W > 0$ almost everywhere on the unit circle, with $\int W(\theta) d\theta/(2\pi) < \infty$ and $\int |\log W(\theta)| d\theta < \infty$. Put

$$e_m = \min_{q(0)=1, \deg q \leq m} \int_{-\pi}^{\pi} |q(e^{i\theta})|^2 W(\theta) \frac{d\theta}{2\pi}, \quad G = \exp \int_{-\pi}^{\pi} \log W(\theta) \frac{d\theta}{2\pi}. \quad (12.1)$$

Reversing coefficients shows that e_m is also the minimum squared norm of a monic polynomial of degree m .

Theorem 12.1 (Szegő minimum principle). *The numbers e_m are nonincreasing and converge to $G > 0$.*

Proof. The admissible spaces in (12.1) increase with m , proving monotonicity. For a polynomial q with $q(0) = 1$, Jensen's formula gives $\int \log |q(e^{i\theta})| d\theta/(2\pi) \geq 0$. The arithmetic–geometric mean inequality applied to $|q|^2 W$ then gives $\|q\|_W^2 \geq G$. Hence $e_m \geq G$.

For the converse, fix $\varepsilon > 0$ and let $W_\varepsilon = W + \varepsilon$, $G_\varepsilon = \exp \int \log W_\varepsilon d\theta/(2\pi)$. Define the zero-free analytic function in the unit disk

$$D_\varepsilon(z) = \exp \left\{ \frac{1}{4\pi} \int_{-\pi}^{\pi} \frac{e^{i\theta} + z}{e^{i\theta} - z} \log W_\varepsilon(\theta) d\theta \right\}.$$

The real part of the kernel is the Poisson kernel. Consequently $D_\varepsilon(0) = \sqrt{G_\varepsilon}$, $|D_\varepsilon(e^{i\theta})|^2 = W_\varepsilon(\theta)$ almost everywhere, and $|D_\varepsilon(z)| \geq \sqrt{\varepsilon}$. The standard radial boundary assertion here follows from the Poisson boundary theorem and the Hardy-space outer-function construction. Equivalently, it can be obtained by first truncating $\log W_\varepsilon$ and passing to its Poisson integrals.

Thus $f(z) = D_\varepsilon(0)/D_\varepsilon(z)$ is bounded analytic, $f(0) = 1$, and its boundary values satisfy

$$|f(e^{i\theta})|^2 = G_\varepsilon / W_\varepsilon(\theta).$$

For fixed $r < 1$, the Taylor polynomials of $f(rz)$ converge uniformly on $|z| \leq 1$ and have constant term one. They are admissible competitors. As $r \uparrow 1$, boundedness of f and integrability of W permit dominated convergence, giving

$$\inf_m e_m \leq \int |f|^2 W \frac{d\theta}{2\pi} = G_\varepsilon \int \frac{W}{W + \varepsilon} \frac{d\theta}{2\pi} \leq G_\varepsilon.$$

Finally $\log(W + \varepsilon) \rightarrow \log W$ in integral as $\varepsilon \downarrow 0$. For $0 < \varepsilon \leq 1$, the nonnegative difference is dominated by $\log(W + 1) - \log W$, an integrable function. Hence $G_\varepsilon \rightarrow G$ and the upper bound follows. $\square$

12.1 Reflection coefficients and an exact product

Let $\Phi_m(z)$ be the monic orthogonal polynomial for W and write $\Phi_m^*(z) = z^m \overline{\Phi_m(1/\bar{z})}$. Assume from now on that W is even, so all coefficients of Φ_m are real. Define $\alpha_m = -\Phi_{m+1}(0)$.

Lemma 12.2. For $m \geq 0$,

$$\Phi_{m+1} = z\Phi_m - \alpha_m \Phi_m^*, \quad e_{m+1} = (1 - \alpha_m^2)e_m, \quad |\alpha_m| < 1. \quad (12.2)$$

If $\int W d\theta/(2\pi) = 1$, then

$$e_m = \prod_{j=0}^{m-1} (1 - \alpha_j^2), \quad G = \prod_{j=0}^{\infty} (1 - \alpha_j^2), \quad \sum_{j=0}^{\infty} \alpha_j^2 < \infty. \quad (12.3)$$

Proof. Both Φ_{m+1} and $z\Phi_m$ are orthogonal to $z, z^2, \dots, z^m$. Their difference has degree at most m and therefore is a scalar multiple of Φ_m^* , the one-dimensional orthogonal complement of those m monomials in that space. The constant term gives the scalar $-\alpha_m$. The identity $z\Phi_m = \Phi_{m+1} + \alpha_m \Phi_m^*$ is an orthogonal sum; its norms give $e_m = e_{m+1} + \alpha_m^2 e_m$. All norms are positive, proving $|\alpha_m| < 1$. Iterate and use Theorem 12.1. Since $-\log(1-t) \geq t$ for $0 \leq t < 1$, positivity of the limiting product implies the final assertion. $\square$

13 Exact circle-to-Hankel identities

Lift the measure $\varphi(x) dx$ to the circle by

$$x = \frac{C}{2}(1 + \cos \theta) = \Lambda(z + 2 + z^{-1}), \quad z = e^{i\theta}, \quad W(\theta) = \frac{\pi C}{2} \varphi(x) |\sin \theta|. \quad (13.1)$$

This is an even probability density relative to $d\theta/(2\pi)$. Indeed, its pushforward under $\theta \mapsto x$ is $\varphi(x) dx$. Lemma 11.1, together with the integrability of $\log |\sin \theta|$, verifies the hypotheses of Theorem 12.1. Since the mean of $\log |\sin \theta|$ is $-\log 2$,

$$G = \frac{\pi C}{4} \exp \langle \log \varphi \rangle_C = K/2. \quad (13.2)$$

Jensen's inequality and $\int W = 1$ also give $G \leq 1$.

Lemma 13.1 (Reciprocal and anti-reciprocal minimum norms). For $m \geq 1$, put $a_0 = \Phi_m(0) = -\alpha_{m-1}$. The monic polynomials

$$S_m = \frac{\Phi_m + \Phi_m^*}{1 + a_0}, \quad T_m = \frac{\Phi_m - \Phi_m^*}{1 - a_0}$$

minimize the norm among real monic reciprocal and anti-reciprocal polynomials of degree m , respectively. Their squared norms are

$$\|S_m\|^2 = \frac{2e_m}{1 - \alpha_{m-1}}, \quad \|T_m\|^2 = \frac{2e_m}{1 + \alpha_{m-1}}. \quad (13.3)$$

Proof. The denominators are nonzero by Lemma 12.2. A variation within either monic constrained space has zero leading and constant coefficients and is therefore a polynomial of degree at most $m-1$. It is orthogonal to Φ_m . Reciprocity or anti-reciprocity implies it is also orthogonal to Φ_m^* : on the circle, use $\Phi_m^* = z^m \overline{\Phi_m}$ and $R^* = \pm R$. Thus S_m and T_m are orthogonal to their respective variation spaces and are the unique minimizers.

Moreover, $\|\Phi_m^*\|^2 = e_m$ and $\langle \Phi_m, \Phi_m^* \rangle = a_0 e_m$, since the leading coefficient of Φ_m^* is a_0 . Expanding the squared norms of the sum and difference proves (13.3). $\square$

Define, with the weighted norms from (11.3),

$$U_n = \frac{h_n[1]}{\Lambda^{2n}}, \quad V_n = \frac{h_{n-1}[x(C-x)]}{\Lambda^{2n}} \quad (n \geq 1), \quad (13.4)$$

$$X_n = \frac{h_n[x]}{\Lambda^{2n+1}}, \quad Y_n = \frac{h_n[C-x]}{\Lambda^{2n+1}} \quad (n \geq 0). \quad (13.5)$$

Proposition 13.2 (Four exact norm identities). *One has*

$$U_n = \frac{2e_{2n}}{1 - \alpha_{2n-1}}, \quad V_n = \frac{2e_{2n}}{1 + \alpha_{2n-1}}, \quad (13.6)$$

$$X_n = \frac{2e_{2n+1}}{1 - \alpha_{2n}}, \quad Y_n = \frac{2e_{2n+1}}{1 + \alpha_{2n}}. \quad (13.7)$$

Proof. Every monic reciprocal polynomial of degree $2n$ has the form $z^n p(x)/\Lambda^n$, where p is monic of degree n . Every monic anti-reciprocal polynomial of degree $2n$ has the form $z^{n-1}(z^2 - 1)q(x)/\Lambda^{n-1}$, with q monic of degree $n - 1$. These statements follow by expressing symmetric Laurent polynomials as polynomials in $z + z^{-1}$. Their minimum norms are U_n and V_n , because

$$|z - z^{-1}|^2 = 4 \sin^2 \theta = \frac{x(C - x)}{\Lambda^2}.$$

For degree $2n + 1$, the corresponding forms are $z^n(z + 1)p(x)/\Lambda^n$ and $z^n(z - 1)p(x)/\Lambda^n$. Use $|z + 1|^2 = x/\Lambda$ and $|z - 1|^2 = (C - x)/\Lambda$ to obtain X_n, Y_n . Now apply Lemma 13.1. $\square$

14 Monotone envelopes and the proof of the main theorem

Theorem 14.1 (Interleaved Hankel upper envelopes). *Set $E_0 = 2$ and*

$$E_{2n} = \frac{2U_n V_n}{U_n + V_n} \quad (n \geq 1), \quad E_{2n+1} = \frac{2X_n Y_n}{X_n + Y_n} \quad (n \geq 0). \quad (14.1)$$

Then

$$E_m = 2e_m, \quad E_0 \geq E_1 \geq E_2 \geq \dots \downarrow K. \quad (14.2)$$

In particular, every E_m is a finite expression in moments that is an upper bound for the integral constant K . The reflection coefficients themselves are given exactly by

$$\alpha_{2n-1} = \frac{U_n - V_n}{U_n + V_n}, \quad \alpha_{2n} = \frac{X_n - Y_n}{X_n + Y_n}. \quad (14.3)$$

Proof. Substitute (13.6) and (13.7) into the harmonic means in (14.1). The result is $2e_m$. Theorem 12.1 and (13.2) give the limit and monotonicity. Subtracting and adding the paired identities gives (14.3). $\square$

All entries involved are explicit linear combinations of Apéry numbers: for example, the $x(C - x)$ -modified moment of degree k is $CA_{k+1} - A_{k+2}$. Thus the theorem uses only the field $\mathbb{Q}(\sqrt{2})$ for its finite expressions. In direct determinant notation its even part is

$$E_{2n} = \frac{2D_n[1]D_{n-1}[x(C - x)]}{\Lambda^{2n}(D_n[1]D_{n-2}[x(C - x)] + D_{n-1}[1]D_{n-1}[x(C - x)])}. \quad (14.4)$$

Our convention $D_{-1}[P] = 1$ includes $n = 1$. The first simple certificate is

$$K \leq E_1 = 2 \left(1 - \left(\frac{10}{C} - 1 \right)^2 \right) = 81120\sqrt{2} - 114720. \quad (14.5)$$

The radical expression is positive; subtracting its rounded decimal terms in low precision is not a reliable way to evaluate it.

Corollary 14.2 (Exact product and separation of fluctuations). *The constant and normalized norms satisfy*

$$K = 2 \prod_{j=0}^{\infty} (1 - \alpha_j^2), \quad (14.6)$$

$$\log \frac{E_m}{K} = \sum_{j=m}^{\infty} -\log(1 - \alpha_j^2), \quad (14.7)$$

$$\frac{U_n}{K} = \frac{1}{1 - \alpha_{2n-1}} \exp \left\{ \sum_{j=2n}^{\infty} -\log(1 - \alpha_j^2) \right\}. \quad (14.8)$$

Furthermore,

$$\sum_{n \geq 1} \left(\frac{U_n}{E_{2n}} - 1 \right)^2 < \infty. \quad (14.9)$$

Proof. The first two identities follow from (12.3) and $K = 2G$. The third is (13.6). Since $\alpha_j \rightarrow 0$, $U_n/E_{2n} - 1 = \alpha_{2n-1}/(1 - \alpha_{2n-1})$ has a square-summable tail; finitely many initial terms cause no problem. $\square$

Proof of Theorem 10.1. Lemma 11.1 proves the integral assertion. By (13.6), $e_{2n} \rightarrow K/2$ and $\alpha_{2n-1} \rightarrow 0$, so $U_n \rightarrow K$. Equations (13.4) and (11.4) yield (10.3). Summing $\log h_j = 2jL + \log K + o(1)$ gives (10.4); the sum of a sequence converging to zero is $o(n)$. The identity $(1 + 1/\sqrt{2})^4 = \Lambda$ proves (10.5), with the extra factor Λ caused by the indexing of the plotted ratio. Finally $D_{n+1}D_{n-1}/D_n^2 = h_{n+1}/h_n \rightarrow \Lambda^2$. $\square$

Remark 14.3 (What an upper envelope does not give). Monotonicity supplies an exact upper bound for K , but not a certified error estimate from one finite moment table. Equation (14.7) identifies the missing tail. The numerical values of the envelopes below are evaluated in high precision, not by directed interval rounding.

15 All fixed nonnegative polynomial modifications

Let $P(x) = p_d \prod_{j=1}^d (x - \zeta_j)$ be a nonzero real polynomial that is nonnegative on $[0, C]$. Its zeros on this interval are permitted. Set

$$\mathcal{G}_C(P) = \exp \langle \log P \rangle_C. \quad (15.1)$$

Zeros are understood in the integrable logarithmic sense.

Theorem 15.1 (Polynomial perturbation constants). *For each fixed such P ,*

$$h_n[P] \sim K \mathcal{G}_C(P) \Lambda^{2n}, \quad (15.2)$$

$$\log D_n[P] = n(n+1)L + (n+1) \log(K \mathcal{G}_C(P)) + o(n), \quad (15.3)$$

$$\frac{h_n[P]}{h_n[1]} \rightarrow \mathcal{G}_C(P), \quad \left(\frac{D_n[P]}{D_n[1]} \right)^{1/(n+1)} \rightarrow \mathcal{G}_C(P). \quad (15.4)$$

Proof. The density $P\varphi$ is integrable and positive almost everywhere. The logarithm of each polynomial factor has an integrable singularity against the arcsine measure, including at the endpoints. Consequently it satisfies (11.1) with $\log(P\varphi) = \log P + \log \varphi$. Repeat the circle minimum argument. A probability normalization is unnecessary for the norm limit: it only changed e_0 in the product formula. The new constant is exactly $K \mathcal{G}_C(P)$. The product identity (11.4) and cancellation prove the rest. $\square$

15.1 The constant is explicit from the roots

Define the exterior conformal parameter

$$\Psi_C(z) = \frac{z - C/2 + \sqrt{z(z - C)}}{2\Lambda}, \quad z \notin [0, C], \quad (15.5)$$

where the square root is chosen to be $z - C/2 + O(1/z)$ at infinity. Then $|\Psi_C(z)| > 1$. On the interval, use either boundary value, of modulus one, when only its modulus occurs.

Proposition 15.2 (Root formula). *For the polynomial above,*

$$\mathcal{G}_C(P) = |p_d| \Lambda^d \prod_{j=1}^d |\Psi_C(\zeta_j)|. \quad (15.6)$$

Proof. Let $w = \Psi_C(\zeta)$, so $\zeta = \Lambda(w + 2 + w^{-1})$ and $|w| \geq 1$. For $|z| = 1$,

$$\Lambda(z + 2 + z^{-1}) - \zeta = \Lambda z^{-1}(z - w)(z - w^{-1}).$$

The circle average of $\log |z - a|$ is $\log \max(1, |a|)$ by Jensen's formula (or its elementary factorization proof). Hence $\langle \log |x - \zeta| \rangle_C = \log \Lambda + \log |w|$. This remains true for $|w| = 1$, since the boundary logarithmic singularities are integrable. Add over the roots and the leading coefficient. $\square$

A particularly useful consequence is the following location-independence: if all roots of P lie in $[0, C]$, then

$$\mathcal{G}_C(P) = |p_d| \Lambda^d. \quad (15.7)$$

Thus a monic square $P(x) = \prod_{j=1}^r (x - t_j)^2$, with fixed $t_j \in [0, C]$, changes the exact linear term in $\log D_n$ by $2r(n+1) \log \Lambda$, independently of the root locations. This does not say that the lower-order terms are independent of those locations.

For fixed integers $r, s \geq 0$,

$$P(x) = x^r (C - x)^s \implies \mathcal{G}_C(P) = \Lambda^{r+s}. \quad (15.8)$$

For an exterior real point $t > C$, $P(x) = t - x$ gives $\mathcal{G}_C(P) = \Lambda \Psi_C(t)$. These are finite algebraic expressions when the roots are algebraic. All assertions are for a fixed polynomial. No uniformity for growing degree or moving roots is implied.

16 Fixed strides and shifts

For fixed integers $m \geq 1$ and $r \geq 0$, consider

$$H_n^{(m,r)} = \det(A_{m(i+j)+r})_{0 \leq i,j \leq n}. \quad (16.1)$$

This is Part I's (4.1) with a shift that does not depend on n . Part I already controls these determinants at the leading scale (Theorem 4.1, and for fixed r the remark after its proof: $\log H_n^{(m,r)} = n(n+1) \log(C^m/4) + O_{m,r}(n)$). Their norm constants can also be identified. Put $B_m = C^m$, $\Lambda_m = B_m/4$, and, with w_m the pushed-forward density of Part I's (4.10),

$$w_m(y) = \frac{1}{m} y^{1/m-1} \varphi(y^{1/m}), \quad \mathcal{K}_m = \frac{\pi B_m}{2} \exp \langle \log w_m \rangle_{B_m}. \quad (16.2)$$

Theorem 16.1 (Sharp fixed-stride constants). *For every fixed m, r as above,*

$$\frac{H_n^{(m,r)}}{H_{n-1}^{(m,r)}} \sim \mathcal{K}_m \Lambda_m^{r/m} \Lambda_m^{2n}, \quad (16.3)$$

$$\log H_n^{(m,r)} = n(n+1) \log \Lambda_m + (n+1) \left(\log \mathcal{K}_m + \frac{r}{m} \log \Lambda_m \right) + o(n). \quad (16.4)$$

In particular, dividing the consecutive norm ratio for shift r by the one for shift zero has limit $\Lambda_m^{r/m}$.

Proof. The substitution $y = x^m$ gives

$$A_{mk+r} = \int_0^{B_m} y^k y^{r/m} w_m(y) dy.$$

Near zero, $\log w_m(y) = (1/m - 1) \log y + \log \log(1/y) + O(1)$. At B_m it has the same square-root endpoint order as before, and at c^m the same finite-order Frobenius behavior after an analytic change of coordinate. The proof of Lemma 11.1 therefore applies. The further factor $y^{r/m}$ preserves logarithmic integrability. The circle argument yields a norm constant multiplied by $\exp((r/m) \langle \log y \rangle_{B_m}) = \Lambda_m^{r/m}$. Summing logarithms proves (16.4). $\square$

This theorem deliberately keeps m and r fixed. It does not replace Part I's separate varying-shift Theorem 4.1 when r is proportional to n . An exact linear correction in that varying regime remains a further problem. [Added 2 October 2026, batch 77P5: Part III answers this for stride $m = 1$ and shifts $r = sN$ with s in a fixed compact subset of (s_*, ∞) , $s_* = 3\sqrt{2}/4 - 1$, where N is the matrix order: the linear coefficient is $B(s)$ of (22.11), and Corollary 22.2 also gives the power $N^{-1/24}$ and the constant $M(s)$ (Theorem 28.1 adds the first $1/N$ correction). Section 27 records how the linear coefficient changes when $r = \lfloor \rho N \rfloor$ or $r = \lfloor \rho n \rfloor$ with $N = n + 1$. Strides $m \geq 2$ and shifts with $s \leq s_*$, including the approach to the cusp collision at s_* , remain open.]

Remark 16.2 (Relation to Part I). The quadratic term $n(n+1) \log \Lambda_m = n(n+1) \log(C^m/4)$ in (16.4) is Part I's Theorem 4.1 at $\rho = 0$ (limit $C^m/4$, which is not $(C/4)^m$), in the sharper form of the remark after its proof. What Theorem 16.1 adds is the linear coefficient $\log \mathcal{K}_m + (r/m) \log \Lambda_m$. For $m = 1$, $r = 0$ it is Theorem 10.1: $w_1 = \varphi$ and $\mathcal{K}_1 = K$.

17 Inverse growth and the integer-index issue

Set

$$B = L + \log K. \quad (17.1)$$

For sufficiently large n , $D_{n+1} > D_n$ by (10.3). Let $F(x)$ be the piecewise-linear interpolation of $\log D_n$ on that increasing tail, and let $X(y)$ be its inverse.

Theorem 17.1 (Two-term inverse-index law). *As $y \rightarrow \infty$,*

$$X(y) = \sqrt{\frac{y}{L}} - \frac{L + \log K}{2L} + o(1). \quad (17.2)$$

For a large threshold T , the first index on the increasing tail with $D_n \geq T$ is exactly $\lceil X(\log T) \rceil$. Equivalently, it has the form

$$\left\lceil \sqrt{\frac{\log T}{L}} - \frac{L + \log K}{2L} + \varepsilon(T) \right\rceil, \quad \varepsilon(T) \rightarrow 0. \quad (17.3)$$

Proof. Equation (10.4) gives $\log D_n = Ln^2 + Bn + o(n)$. Linear interpolation changes Lx^2 by at most $L/4$, and interpolating an $o(n)$ remainder still gives $o(x)$. Thus $F(x) = Lx^2 + Bx + o(x)$. First $X(y) \sim \sqrt{y/L}$. Completing the square gives

$$y = L \left(X(y) + \frac{B}{2L} \right)^2 + o(X(y)),$$

which implies (17.2) after division by a quantity of order $\sqrt{y}$. The ceiling statement follows directly from monotonicity and the definition of the interpolation. $\square$

Numerically,

$$L \approx 2.139199986958, \quad B \approx 1.682590155276, \quad -\frac{B}{2L} \approx -0.393275562251.$$

Thus the constant missing from the $O(n)$ estimate (1.6) of Part I produces an explicit constant shift in the inverse. One must not remove the ceiling in (17.3) and claim an $o(1)$ approximation to an integer-valued staircase. Nor may one drop the small term inside the ceiling near an integer without a separate error bound.

18 A conditional transfer theorem for finer transseries

The exact formula (14.8) separates the remaining analytic problem from subsequent asymptotic algebra. Once the reflection coefficients are understood, the determinant expansion follows by tail summation. The next theorem is an unconditional implication, but its hypothesis is *not proved here for the Apéry density*.

Theorem 18.1 (Two slow scales of reflection coefficients). *For a lifted probability density as above, suppose that some real a, b, d, e satisfy, as $j \rightarrow \infty$,*

$$\alpha_j = \frac{a + b(-1)^j}{j} + \frac{d + e(-1)^j}{j \log j} + O\left(\frac{1}{j \log^2 j}\right). \quad (18.1)$$

Define

$$\chi = \frac{a^2 + b^2 + a - b}{2}, \quad \xi = ad + be + \frac{d - e}{2}. \quad (18.2)$$

Then

$$\log \frac{U_n}{K} = \frac{\chi}{n} + \frac{\xi}{n \log n} + O\left(\frac{1}{n \log^2 n}\right), \quad (18.3)$$

and there exists a finite constant $\mathcal{A} > 0$ such that

$$D_n = \mathcal{A} K^{n+1} \Lambda^{n(n+1)} n^\chi (\log n)^\xi \left(1 + O\left(\frac{1}{\log n}\right) \right). \quad (18.4)$$

Proof. Square (18.1). The nonalternating terms are

$$\frac{a^2 + b^2}{j^2} + \frac{2(ad + be)}{j^2 \log j} + O\left(\frac{1}{j^2 \log^2 j}\right).$$

The remaining displayed terms alternate and have tail $O(m^{-2})$ from $j = m$ onward, by summation by parts. The elementary integral estimates are

$$\sum_{j=m}^{\infty} j^{-2} = m^{-1} + O(m^{-2}),$$

$$\sum_{j=m}^{\infty} \frac{1}{j^2 \log j} = \frac{1}{m \log m} + O\left(\frac{1}{m \log^2 m}\right).$$

Also $-\log(1 - \alpha_j^2) = \alpha_j^2 + O(j^{-4})$. Therefore

$$\sum_{j=m}^{\infty} -\log(1 - \alpha_j^2) = \frac{a^2 + b^2}{m} + \frac{2(ad + be)}{m \log m} + O\left(\frac{1}{m \log^2 m}\right).$$

In (14.8), set $m = 2n$ and add $-\log(1 - \alpha_{2n-1}^2)$. Replacing $2n - 1$ by $2n$ costs $O(n^{-2})$ in the principal terms, and replacing $\log(2n)$ by $\log n$ costs $O(1/(n \log^2 n))$. The resulting coefficients are exactly (18.2).

Now sum (18.3). The differences between the partial sums of $1/n$ and $\log n$, and of $1/(n \log n)$ and $\log \log n$, converge to constants. The remainder series converges absolutely, with tail $O(1/\log n)$. Exponentiation and (11.4) prove (18.4); finite initial terms are absorbed into $\mathcal{A}$. $\square$

18.1 A next inverse term, including the lattice correction

There is a subtle but useful distinction between inverting a smooth asymptotic model and inverting the polygonal interpolation of a sequence. Let $\tau(u) = \{u\}(1 - \{u\})$, a continuous, Lipschitz, period-one function.

Proposition 18.2 (Conditional inverse with lattice sector). *Suppose*

$$\log D_n = Ln^2 + Bn + \chi \log n + \xi \log \log n + C_0 + o(1), \quad L > 0. \quad (18.5)$$

With the same polygonal inverse X as before, put $s = \sqrt{y/L}$. Then

$$\begin{aligned} X(y) = s - \frac{B}{2L} \\ + \frac{B^2/(4L) - \chi \log s - \xi \log \log s - C_0 - L\tau(s - B/(2L))}{2Ls} + o(s^{-1}). \end{aligned} \quad (18.6)$$

For the inverse of the smooth expression in (18.5), the same formula holds with the τ term omitted.

Proof. For $x = n + t$, $0 \leq t \leq 1$, linear interpolation of n^2 is exactly $x^2 + t(1 - t)$. Interpolation of the logarithmic terms differs from their smooth values by $o(1)$. Thus

$$F(x) = Lx^2 + Bx + \chi \log x + \xi \log \log x + C_0 + L\tau(x) + o(1).$$

Substitute $x = s - B/(2L) + v/s$. The quadratic and linear terms become $y - B^2/(4L) + 2Lv + o(1)$ when $v = O(\log s)$. All logarithms may be evaluated at s to this accuracy. Since τ is Lipschitz and periodic, it may be evaluated at $s - B/(2L)$, even across integer boundaries, with error $o(1)$. Equating the residual to zero gives the displayed value of v . A residual $o(1)$ in F changes its inverse by $o(s^{-1})$, since the local slopes are $2Ls + O(1)$. $\square$

Theorem 18.1 supplies (18.5) with $B = L + \log K$ and $C_0 = \log \mathcal{A} + \log K$, whenever its reflection-coefficient hypothesis is verified. This isolates a reusable asymptotic-inversion step and prevents a spurious constant term caused by ignoring interpolation.

19 Computational checks and numerical results

The package separates exact finite checks from numerical asymptotics. Neither substitutes for the analytic proof.

19.1 Exact algebraic checks

The script `code/02-szego-verify_exact.py` uses only Python's standard library. It implements arithmetic in $\mathbb{Q}(\sqrt{2})$ using pairs of rational numbers and exact sign tests. It verifies the following at the included cutoff:

1. The binomial definition and recurrence for the required Apéry moments, and seven independent integer Bareiss determinants, D_0 through D_6 .
2. Four families of minimum norms through degree six, obtained by exact Gram elimination, and fourteen circle minimum norms from an independently constructed Toeplitz matrix of Chebyshev moments.
3. Every applicable identity $E_m = 2e_m$, all thirteen intervening norm recurrences, and exact nonincrease of the interleaved envelopes.

The run completed with every assertion passing. For example,

$$E_1 = 81120\sqrt{2} - 114720, \quad E_2 = \frac{12989952}{5} - 1837056\sqrt{2}.$$

The output is recorded in `data/02-szego-exact_checks.json`. These are normalization and algebra checks at finite indices, not a formal verification of the infinite-dimensional argument.

19.2 High-precision moment matrices

The C++ program `code/02-szego-moment_norms.cpp` uses GMP arithmetic. It first forms the exact integer Apéry moments and scales them to $M_k = A_k/C^k$, the moments of a measure on $[0, 1]$. The four moment lists used are

$$M_k, \quad M_{k+1}, \quad M_k - M_{k+1}, \quad M_{k+1} - M_{k+2}.$$

Symmetric elimination yields every leading minimum norm without taking square roots or separately recomputing each determinant. Scaling back gives U_n, V_n, X_n, Y_n .

The calculation was run through $n = 200$ at both 3072-bit and 4096-bit precision. All emitted 70-significant-digit values agreed between the two runs. This checks numerical stability at those working precisions; it is not interval certification. The program rejects nonpositive pivots and violations of the proved interleaved monotonicity.

n	U_n	E_{2n}	E_{2n+1}	ΛU_n
10	.633485989362	.666306510245	.665096192052	5.379968887979
20	.631012979942	.650723277428	.650526912374	5.358966507555
50	.632544044029	.640453417925	.640419013866	5.371969284714
100	.632828523882	.637019913211	.637012787034	5.374385269888
150	.632966981321	.635846665253	.635843846206	5.375561139169
200	.633051776322	.635253009843	.635251548628	5.376281272648

The final column is the OEIS-style ratio with index $n - 1$. The difference between the $n = 200$ ratio and its limit is still about 0.00319, despite the apparent flattening of parts of the curve.

[Added 1 October 2026, batch 73O1: the two runs are shipped as `data/02-szego-norms.csv` (4096 bits) and `data/02-szego-norms_3072.csv`; since every emitted digit agreed, the two files are byte-identical. At intake, the value $U_{200} = 0.633051776322250247495064895880637 \dots$ of this table was compared with the column `normalized_norm` of Part I's `data/convergence.csv`, computed from the exact integer determinants of Section 7: the two pipelines agree in all 33 digits that Part I prints. Independent 900-digit Hankel eliminations at $n = 10, 20, 30$ also agreed with U_n and E_{2n} of the shipped CSV in the 15 significant digits compared.]

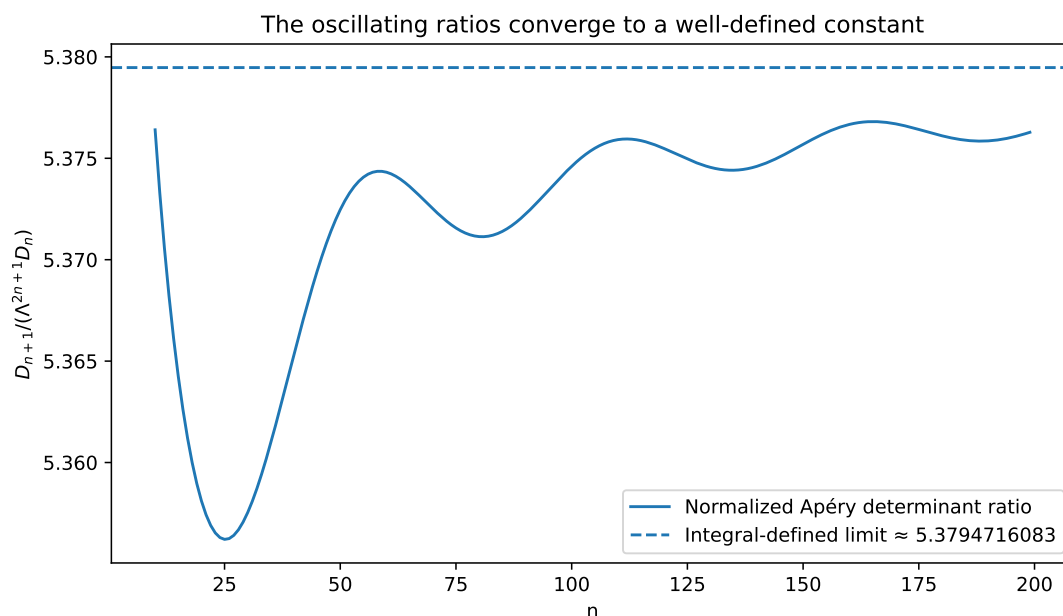


Figure 2: High-precision evaluation of the normalized determinant ratios. The horizontal line is the independently evaluated integral constant ΛK , not a fitted limit. Theorem 10.1 proves convergence; the numerical curve alone does not.

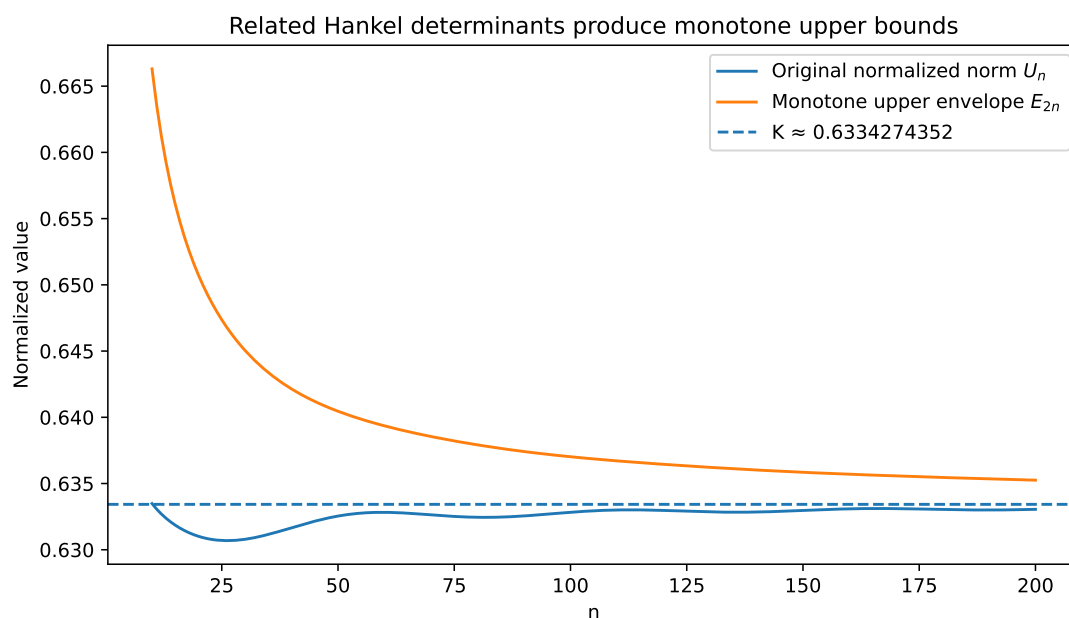


Figure 3: The original normalized norm and the even subsequence of the monotone upper envelopes. The omitted odd envelopes lie between consecutive even ones by Theorem 14.1.

19.3 Independent density quadrature

The script `code/02-szego-density_constant.py` evaluates the density by hypergeometric continuation, with endpoint expansions to avoid catastrophic cancellation. It integrates after $x = C \sin^2(t/2)$, so the arcsine measure becomes dt/π . It gives

$$\begin{aligned}\langle \log \varphi \rangle_C &\approx -4.433686885050, \\ K &\approx 0.633427435172, \\ \Lambda K &\approx 5.379471608281.\end{aligned}$$

As an independent normalization check, quadrature of the density reproduced A_0 through A_4 , namely 1, 5, 73, 1445, 33001. At 40 decimal working digits, the largest reported absolute discrepancy was below 10^{-28} . Extra digits written to the JSON output are diagnostic; there is no rigorous decimal error enclosure in this package.

19.4 An inverse check

At $y = \log D_n$, the exact polygonal inverse is $X(y) = n$. Using only the proved two-term formula, its numerical error $\sqrt{\log D_n/L - B/(2L)} - n$ is approximately

n	20	50	100	200
error	−0.00279410	−0.00157978	−0.00093813	−0.00055161.

This is consistent with the $o(1)$ error proved in Theorem 17.1; no stronger rate is inferred from the table.

20 Further research questions and a concrete route forward

The proved statements identify a precise next analytic task rather than assuming a generic inverse-power expansion. They re-scope the continuation proposed at the end of Part I (Section 8): the limit of h_n/Λ^{2n} and the linear term of $\log D_n$ are now determined (Theorem 10.1); the prefactor, the oscillation and the varying-shift linear term are not. [Added 2 October 2026, batch 77P5: Part III determines the varying-shift linear term for stride one and $s = r/N > s_* = 3\sqrt{2}/4 - 1$, together with the power of N , the constant and the first correction (Corollary 22.2, Theorem 28.1). The prefactor of D_n itself ($s = 0$) and the oscillation remain open: Part III's prefactor is proved only on compact subsets of (s_*, ∞) and must not be evaluated at $s = 0$.]

20.1 Quantitative reflection-coefficient asymptotics

Determine the first terms of α_j for the lifted Apéry density. Its two endpoint behaviors and its interior singularity must all be retained. Can one prove an expansion of the form (18.1), with explicit a, b, d, e and a uniform remainder? This would immediately unlock Theorem 18.1, including a polynomial/logarithmic prefactor for D_n and the next inverse terms. A sharper remainder would propagate through the exact tail formula (14.8) to higher orders.

20.2 The oscillatory sector and its frequency

Let

$$\theta_* = \arccos(2c/C - 1) = \pi - 2 \arcsin(1/C).$$

The interior singularity maps to circle points $e^{\pm i\theta_*}$. Determine whether the leading oscillatory correction is governed by $e^{\pm 2in\theta_*}$, and calculate its amplitude, phase, and decay exponent. The corresponding candidate period in n is $\pi/(2 \arcsin(1/C)) \approx 53.3531$. The geometry suggests this frequency, but this report does *not* prove that it is the leading oscillatory term. The visible oscillations make this a more informative target than an unqualified fit to powers of $1/n$.

20.3 A full Hankel prefactor and logarithmic endpoint effects

Does there exist $\mathcal{A} > 0$ and a rigorously determined exponent χ with $D_n \sim \mathcal{A}K^{n+1}\Lambda^{n(n+1)}n^\chi$, or is an additional power of $\log n$ required? The logarithmic singularity of φ at zero means that one must verify, not merely assume, the hypotheses of a smooth-background Fisher–Hartwig theorem. The general determinant asymptotics in [6] provide a relevant comparison framework, but are not invoked here to claim an unproved prefactor.

20.4 Certified constants and effective upper-envelope tails

Construct rigorous interval enclosures for K by combining density quadrature with explicit endpoint and interior remainder bounds. Alternatively, obtain an effective bound on $\sum_{j \geq m} -\log(1 - \alpha_j^2)$. The exact finite envelopes already supply algebraic upper bounds; a controlled tail would turn them into two-sided numerical certificates. Closed-form simplifications of K in terms of special values are also worth investigating, but no such simplification is asserted.

20.5 Growing shifts and moving polynomial zeros

For $r_n/n \rightarrow \rho > 0$, Part I (Theorem 4.1) determines the leading quadratic term for $H_n^{(m, r_n)}$. Identify the next linear term as a function of ρ , including regimes where a moving equilibrium-support endpoint approaches the density’s interior singularity. Separately, extend Theorem 15.1 uniformly to moving roots or growing polynomial degree. The fixed-polynomial result alone does not justify those limits. [Added 2 October 2026, batch 77P5: Part III identifies the next linear term, and also the power of N , the constant and the first $1/N$ correction, for stride one and $\rho > s_* = 3\sqrt{2}/4 - 1$; there the moving soft edge $a(\rho)$ stays a positive distance to the right of the interior singularity, which sits at $t_c = C^{-2}$ in the scaled variable $t = x/C$ (Section 23). The regime named here, in which the moving endpoint approaches the interior singularity, is exactly $\rho \downarrow s_*$; its double-scaling analysis remains open (Section 30, item 1). Strides $m \geq 2$ and the extension to moving roots or growing degree are not addressed.]

20.6 Formalization and reuse for other moment sequences

The exact determinant/Gram identities and the reciprocal-polynomial reductions form a finite-algebraic formalization target. The next layer is the circle minimum principle and the transfer of the Szegő condition. Once these are formalized, the Apéry application has a clean dependency on an independently verified moment-density theorem. For other OEIS moment sequences, the same pipeline needs three inputs: a precise positive density, its interval support, and logarithmic integrability against the arcsine measure. The envelope construction then uses moments alone.

Part III. Full equivalents for proportionally shifted Apéry Hankel determinants

Explicit asymptotic constants and phase-aware inversion above the support cusp collision

Research report prepared with ChatGPT, dated 2 October 2026; added to this report on 2 October 2026 in batch 77P5 of ProveIt's incoming-report intake.

Part III was written against Parts I and II as printed above (repository commit 946b6c762). For stride one and shifts above a critical proportion, it answers the question that both earlier Parts leave open: the linear term of $\log \det(A_{r+i+j})$ when the shift r is proportional to the matrix order (Section 16, the opening of Section 20, and its subsection on growing shifts). It also gives the power of the order, the constant and the first $1/N$ correction. Section 21.2 records the provenance and fixes the notation of the addition. Read Table 5 first: the matrix order in Part III is N , not $n + 1$.

Summary (the manuscript's abstract). Let $H_N(r) = \det(A_{r+i+j})_{0 \leq i,j < N}$, where A_k is the Apéry number associated with $\zeta(3)$. For $s = r/N$ in a fixed compact subset of $(3\sqrt{2}/4 - 1, \infty)$, we prove

$$H_N(r) = M(s)N^{-1/24} \exp\{N^2 A(s) + NB(s)\}(1 + O(N^{-1})).$$

The functions A, B and the positive constant M are explicit in terms of Edgar's moment density, a one-dimensional integral, an exponentially convergent cosine series and Barnes' G -function. An equivalent form retains an exact Selberg determinant. The proof adapts the Charlier–Gharakhloo determinant-ratio analysis to a logarithmically confining potential; quantitative remote-jump and differential-identity tail estimates supply the needed extension. We also derive an explicit first $1/N$ correction with relative $O(N^{-2})$ error, retain all bounded shift phases and derive asymptotic inverse thresholds on rational residue classes, including the error inside the ceiling. The cusp collision and the full prefactor of the unshifted sequence A228143 are not settled here.

The principal result. Put $C = 17 + 12\sqrt{2}$ and $s_* = 3\sqrt{2}/4 - 1 = 0.06066 \dots$. Uniformly for $s = r/N$ in every fixed compact subset of (s_*, ∞) ,

$$\det(A_{r+i+j})_{0 \leq i,j < N} = M(s) N^{-1/24} e^{N^2 A(s) + NB(s)} \left(1 + \frac{c_{\text{full}}(s)}{N} + O(N^{-2}) \right),$$

with A, B, M of (22.10)–(22.12) and c_{full} of (28.4). The quadratic coefficient $A(s) = (1+s) \log C - \Psi(s)$ is Part I's limit (Remark 21.1); the linear coefficient $B(s)$, the power $N^{-1/24}$, the constant $M(s) > 0$ and $c_{\text{full}}(s)$ are new. The statement says nothing about $s \leq s_*$: in particular it does not apply to D_n ($s = 0$).

Status. This is an AI-assisted mathematical research report, not a peer-reviewed publication or a proof-assistant formalization. Numerical checks in the accompanying replay are reproducible but are not interval certificates.

21 Scope and prior results

The Apéry numbers are Part I's A_k of (1.1). Part III writes them A_k , in sans-serif, because the italic letter A denotes the quadratic coefficient $A(s)$ of (22.10). As in both earlier Parts, $C = 17 + 12\sqrt{2} = (1 + \sqrt{2})^4$. We study a shift that is constant across each matrix but grows with its order:

$$H_N(r) = \det(A_{r+i+j})_{0 \leq i,j < N}, \quad N \geq 1, \quad r \in \mathbb{Z}_{\geq 0}. \quad (21.1)$$

The original OEIS sequence A228143 uses $D_n = H_{n+1}(0)$. In the notation of Part I's (4.1), $H_N(r) = H_{N-1}^{(1,r)}$: the matrix of $H_n^{(m,r)}$ has order $n + 1$, that of $H_N(r)$ has order N . Parts I and II already establish the quadratic logarithmic law of D_n (Theorem 1.1), its Szegő norm-ratio constant (Theorem 10.1), fixed shifts and strides (Theorems 4.1 and 16.1), and a two-term inverse-index law (Section 17). In particular, with $\Lambda = C/4$ and the constant K of (10.2),

$$\log D_n = n(n+1) \log \Lambda + (n+1) \log K + o(n).$$

That result does not determine the multiplicative prefactor of D_n . The present Part addresses the different, positive-proportional-shift regime. Part II's Section 20 explicitly leaves its linear coefficient and the support/cusp collision open (in its opening paragraph and in its subsection on growing shifts and moving polynomial zeros). Here the equilibrium support lies strictly to the right of the cusp, and we obtain the linear term, the power $N^{-1/24}$ and the full positive leading constant.

The general methods and the moment density are established mathematics. Our contribution is this application with its explicit normalization, confinement adaptation, compact-uniform remainder and shift-phase consequences. The comparison with Parts I and II is dated 2 October 2026. It is not a universal priority claim.

21.1 Relation to Parts I and II

Remark 21.1 (The quadratic term is Part I's). For $s > 0$, the coefficient (22.10) is

$$A(s) = (1 + s) \log C - \Psi(s),$$

with Part I's Ψ of (4.2). This is the limit (4.3) of Part I's Theorem 4.1 at stride $m = 1$ and $\rho = s$; the change of matrix order from $n + 1$ to N does not affect a limit at scale N^2 . The N^2 term of Corollary 22.2 therefore re-proves Part I's growth law on compact subsets of (s_*, ∞) . It is not claimed as new, and the manuscript credits it. Part I's theorem is in other respects more general: it covers every stride $m \geq 1$ and every $\rho \in [0, \infty)$, including $\rho \leq s_*$. What Part III adds, for $m = 1$ and $s > s_*$, is everything below the quadratic term: the exact Selberg normalization (Theorem 22.1), the linear coefficient $B(s)$, the power $N^{-1/24}$, the constant $M(s)$ (Corollary 22.2), the first correction (Theorem 28.1), eventual monotonicity of the floor-shifted sequence, the rational-residue inverse thresholds (Corollary 27.1) and the next inverse term.

The tempting false reading is to let $s \downarrow 0$ in Corollary 22.2 and compare with Part II's $\log D_n = n(n+1) \log \Lambda + (n+1) \log K + o(n)$. The estimates of Part III are uniform only on compact subsets of (s_*, ∞) , and the manuscript states that substituting $s = 0$ into the prefactor is invalid (the remark after Corollary 22.2). No relation between $B(0+)$, $M(0+)$ and Part II's K is asserted.

21.2 Provenance and conventions of Part III

Part III is built from one manuscript, number 10 of batch 77 (cluster P5): *Full equivalents for proportionally shifted Apéry Hankel determinants* (2 October 2026). It was delivered as `apery-shifted-release.zip` in arrival commit 096ee7b87 and placed in commit 4f11bc9c0. Its pin is commit 946b6c762, at which this report consisted of Parts I and II exactly as printed above. Its replay programs, unit tests, expected and recorded outputs and dependency pins are shipped with the prefix `03-shifted-`; the README of this report lists them. The merge made the following choices.

- The manuscript's restated definition of the Apéry numbers (its display (1.1)) is not repeated: Part III cites Part I's (1.1) instead. Every other display, theorem, corollary, lemma, remark, proof, table and question of the manuscript is printed. The manuscript's Section n is Section $n + 20$ here; its abstract is the summary above.

- The manuscript cites Parts I and II as an external repository report pinned at 946b6c762, and Part II's density program as a separate item. Here these citations are references to the numbered statements of Parts I and II and to the shipped file `code/02-szego-density_constant.py`. Its Edgar reference is merged with Part I's bibliography entry. Its other three references are added to the bibliography.
- The quadratic term re-proves Part I's Theorem 4.1 (Remark 21.1). It is not claimed as new.
- The context paragraph after Corollary 27.1, which relates the threshold corollary to the general staircase theorem of the repository's transseries volume, is an editorial addition of the merge.
- File names in the text are the shipped names. The delivered programs still import and read one another under their delivery names (see the README).

Notation. Part III keeps the manuscript's symbols except for the renamings marked $\star$ in Table 5. Where a letter is shared with Parts I and II, the table prints the earlier meaning beside the Part III one. No normalization changed.

Within Part III, $S_N(r, b)$ of (22.9) is the Selberg product (the replay calls $S_N(r, \frac{1}{2}) J_N(r)$) and $\mathcal{S}(u)$ of (28.2) is a cosine energy. In the next inverse term, $k = -\frac{1}{24}$ is the coefficient of $\log N$, not an index.

22 Constants and the full equivalent

Edgar's theorem [3] gives a positive density φ on $[0, C]$ with $A_k = \int_0^C x^k \varphi(x) dx$. Define

$$w(t) = C\varphi(Ct), \quad f(t) = \log \frac{w(t)}{\sqrt{1-t}}, \quad 0 < t < 1. \quad (22.1)$$

The function f is real analytic for $t > C^{-2}$ and extends analytically through $t = 1$. Its endpoint value is

$$f(1) = \log \frac{\sqrt{C}}{2^{5/4}\pi^2}. \quad (22.2)$$

Its behavior elsewhere is material to the proof: it has a Hölder cusp at $t_c = C^{-2}$, while w has logarithmic growth at zero. We do not replace it by a globally analytic background.

For $s > 0$, write

$$a = a(s) = \left(\frac{s}{s+2} \right)^2, \quad p_s(t) = \frac{s+2}{2\pi t} \sqrt{\frac{t-a}{1-t}}, \quad a < t < 1, \quad (22.3)$$

$$t_s(\theta) = \frac{1+a}{2} + \frac{1-a}{2} \cos \theta, \quad \hat{f}_k(s) = \frac{1}{\pi} \int_0^\pi f(t_s(\theta)) \cos(k\theta) d\theta. \quad (22.4)$$

Thus the Fourier convention is $f(t_s(\theta)) = \hat{f}_0 + 2 \sum_{k \geq 1} \hat{f}_k \cos(k\theta)$. Set

$$F(s) = \int_a^1 f(t) p_s(t) dt, \quad (22.5)$$

$$E(s) = \exp \left\{ \frac{\hat{f}_0(s) - f(1)}{4} + \frac{1}{2} \sum_{k \geq 1} k \hat{f}_k(s)^2 \right\}. \quad (22.6)$$

On every fixed compact interval strictly above the critical value

$$s_* = \frac{2}{C-1} = \frac{3\sqrt{2}}{4} - 1 = 0.0606601717798212866 \dots, \quad (22.7)$$

the series in (22.6) is exponentially convergent, uniformly in s . The same is true after any fixed number of parameter derivatives.

Table 5: Symbols of Part III that Parts I and II also use.

Symbol	Parts I and II	Part III
$A_k, \mathbf{A}_k, A$	A_k , the Apéry numbers (1.1)	$\mathbf{A}_k$, the same numbers; italic $A(s)$ is the quadratic coefficient (22.10), and $A = A(\rho)$ in (27.6)
N, n, H	D_n and $H_n^{(m,r)}$ (4.1) have matrix order $n+1$	$H_N(r)$ (21.1) has matrix order N : $H_N(r) = H_{N-1}^{(1,r)}$, $D_n = H_{n+1}(0)$. Not $H_N(r) = H_N^{(1,r)}$
s, ρ	$\rho = \lim r_n/n$ (Theorem 4.1); s the power-weight exponent in Section 4	$s = r/N$ exactly; ρ a fixed limit with $r = \rho N - \delta_N$ in Section 27; ρ_V a density in Section 28.3
C, c, Λ	(1.4)	the same; the cusp $x = c$ is $t_c = C^{-2}$ in the scaled variable $t = x/C$
K	Part II's ratio constant (10.2); Part I: a generic constant	Part II's constant (Section 21 only)
$\star K_I$	—	the generic constants of (25.7), (25.8) and Lemma 28.2; the manuscript writes K
E	Part II's envelopes $E_m \downarrow K$ (Section 14)	$E(s)$ of (22.6), an exponential of Fourier data, not an envelope; $\mathcal{E}_N(s)$ is the diagnostic (29.1)
$\star \Theta_j$	—	the coefficient functionals $\Theta_0, \Theta_1, \Theta_2$ of Lemma 28.2; the manuscript writes E_0, E_1, E_2 (its E_0, E_1 are not Part II's envelopes $E_0 = 2$, $E_1 = 81120\sqrt{2} - 114720$)
B	Part II: $B = L + \log K$ (17.1), $B_m = C^m$; Part I: a placeholder	$B(s)$ (22.11); B_δ of (27.6)
$\star \eta_I$	η : Part I's minorant constant (Lemma 2.1)	the tail slopes $\eta_I, \eta_{I,1}$ of (25.5) and Lemma 28.2; the manuscript writes η, η_1
$\star h_{\min}, \ell$	m : the stride (Part III has stride 1)	the lower bound $h \geq h_{\min}$ in (25.3) and the logarithmic exponent ℓ in (25.9); the manuscript writes m for both
φ	Edgar's density	the same density
$\star \vartheta$	—	the angle of the transformed support in Section 25.5; the manuscript writes φ
$\star c_I$	—	the rate in $O(e^{-c_I N})$ (Section 25); the manuscript writes c
h	h_n , the monic norms (3.1)	$h(x) = \exp f((1+x)^{-1})$ (25.3); h_V , a density factor
Q, R	Part I: a polynomial $Q(x)$ and $R = \varphi/Q^2$ (Section 2)	Q_s , the effective potential; R , the small-norm error matrix; $R_s(z)$, a square root
G, D	Part II: $G_\varepsilon, D_\varepsilon(z)$ (Section 12); D_n , the determinants	G , Barnes' function; G_s , the Stieltjes transform (23.1); $D = a'_\varepsilon(0)$ (28.15), not a determinant
J, U	Part I: $J_n(s; L)$ (4.6); Part II: the norms U_n	$J(a, V)$, the endpoint integral (28.11); $U(u)$, an arcsine average (28.1), and the constant U_*
F, L	Part I: the series $\sum D_n z^n$, a length L ; Part II: an interpolation F , $L = \log \Lambda$	$F(s)$ (22.5); $F_a(\theta)$ in Lemma 28.3; $L_s = 4(s+1)/s^2$ (24.5)
a, b, q	Part II: constants of (18.1); a polynomial $q(x)$ (Appendix B)	$a = a(s)$, the soft edge (22.3); $b = \frac{1}{2}$, the hard-edge exponent; $q = 1/(s+1)$ in Section 25.5, q_0 and q_ε elsewhere

Theorem 22.1 (Exact Selberg normalization). *Let I be any fixed compact subset of (s_*, ∞) . Uniformly over all integers $N \geq 1$, $r \geq 0$ with $s = r/N \in I$, as $N \rightarrow \infty$,*

$$H_N(r) = C^{Nr+N(N-1)} S_N(r, \frac{1}{2}) e^{NF(s)} E(s) (1 + O_I(N^{-1})), \quad (22.8)$$

where the reference determinant is the exact finite product

$$S_N(r, b) = \prod_{j=0}^{N-1} \frac{\Gamma(j+1)\Gamma(r+j+1)\Gamma(b+j+1)}{\Gamma(r+b+N+j+1)}. \quad (22.9)$$

For the elementary exponential form, let G denote Barnes' G -function, normalized by $G(1) = 1$ and $G(z+1) = \Gamma(z)G(z)$, and define

$$A(s) = (s+1) \log C + (s+1)^2 \log(s+1) - \frac{s^2}{2} \log s - \frac{(s+2)^2}{2} \log(s+2), \quad (22.10)$$

$$B(s) = \log(2\pi) - \log C + \frac{(s+1) \log(s+1) - (s+2) \log(s+2)}{2} + F(s), \quad (22.11)$$

$$M(s) = \frac{(2\pi)^{1/4} e^{2\zeta'(-1)}}{G(3/2)} \frac{s^{1/12}}{[(s+1)(s+2)]^{1/24}} E(s). \quad (22.12)$$

All three functions are smooth, indeed real analytic, for $s > s_*$, and $M(s) > 0$.

Corollary 22.2 (Full multiplicative equivalent). *Under the assumptions of Theorem 22.1,*

$$H_N(r) = M(s) N^{-1/24} \exp\{N^2 A(s) + N B(s)\} (1 + O_I(N^{-1})), \quad s = \frac{r}{N}. \quad (22.13)$$

Remark 22.3. The compact set I is fixed independently of N . These estimates do not assert uniformity as $s \downarrow s_*$, $s \downarrow 0$ or $s \rightarrow \infty$. In particular, substituting $s = 0$ into the prefactor is invalid. The power $-1/24$ belongs to the present mixed soft/hard regime.

23 Equilibrium measure and the collision

The logarithmic external field on $[0, 1]$ is $V_s(t) = -s \log t$. Its equilibrium measure is (22.3). We give the verification, since it determines both the critical value and the relevant geometry. Let

$$R_s(z) = \sqrt{\frac{z-a}{z-1}}, \quad R_s(z) \rightarrow 1 \quad (z \rightarrow \infty).$$

The candidate Stieltjes transform is

$$G_s(z) = \frac{-s + (s+2)R_s(z)}{2z}. \quad (23.1)$$

It is $z^{-1} + O(z^{-2})$ at infinity. Since $\sqrt{a} = s/(s+2)$, its apparent pole at zero cancels. The boundary jump on $(a, 1)$ is $-2\pi i p_s(t)$, and $G_{s,+}(t) + G_{s,-}(t) = -s/t = V'_s(t)$. These facts prove the mass normalization and the equilibrium equality on the support.

Let Q_s be the effective potential, normalized to zero on $[a, 1]$:

$$Q_s(t) = V_s(t) - 2 \int_a^1 \log |t-u| p_s(u) du + \ell_s.$$

For $0 < t < a$ the chosen branch of R_s is positive, and

$$Q'_s(t) = -\frac{s+2}{t} R_s(t) < 0, \quad Q_s(a) = 0. \quad (23.2)$$

Consequently $Q_s(t) > 0$ throughout the void $(0, a)$. This proves the variational inequality and one-cut regularity. The left edge is a regular soft edge; the fixed right endpoint is a hard edge.

The density cusp is at $t_c = C^{-2}$. The soft edge meets it precisely when $a(s) = C^{-2}$, or $s = 2/(C-1)$, giving (22.7). For $s \in I$ there is a uniform positive distance between the support and the cusp.

24 Exact determinant reduction

Heine's identity and the scaling $x = Ct$ give

$$\begin{aligned} H_N(r) &= \frac{1}{N!} \int_{[0,C]^N} \prod_{i<j} (x_j - x_i)^2 \prod_{j=1}^N x_j^r \varphi(x_j) dx_j \\ &= C^{Nr+N(N-1)} \mathcal{D}_N[t^r(1-t)^{1/2} e^{f(t)}], \end{aligned} \quad (24.1)$$

where

$$\mathcal{D}_N[v] = \frac{1}{N!} \int_{[0,1]^N} \prod_{i<j} (t_j - t_i)^2 \prod_{j=1}^N v(t_j) dt_j.$$

The Selberg integral divided by $N!$ says $\mathcal{D}_N[t^r(1-t)^b] = S_N(r, b)$ for $r, b > -1$. Thus no normalization of a probability density or factorial is left implicit in (22.8).

For each integration variable use the Möbius map

$$x = \frac{1-t}{t}, \quad t = \frac{1}{1+x}. \quad (24.2)$$

The Jacobian contributes $(1+x)^{-2}$ and the squared Vandermonde contributes $(1+x)^{-2(N-1)}$ for each variable. Hence

$$t^r(1-t)^b dt \prod \text{Vandermonde denominators} = x^b(1+x)^{-r-2N-b} dx.$$

The ratio in (24.1) to $S_N(r, b)$ is exactly a ratio of Laguerre-type Hankel integrals on $[0, \infty)$ with

$$V_s(x) = (s+2) \log(1+x), \quad (24.3)$$

$$W_0(x) = -b \log(1+x), \quad W_1(x) = W_0(x) + f((1+x)^{-1}), \quad b = \frac{1}{2}. \quad (24.4)$$

The hard edge is now at zero. The equilibrium measure and its support are

$$d\mu_s(x) = \frac{s}{2\pi(1+x)} \sqrt{\frac{L_s - x}{x}} dx, \quad 0 < x < L_s, \quad L_s = \frac{4(s+1)}{s^2} = a^{-1} - 1. \quad (24.5)$$

After mapping $[0, L_s]$ to $[-1, 1]$ by $u = 2x/L_s - 1$, its analytic coefficient is

$$\psi_s(u) = \frac{sL_s}{4\pi(1+L_s(u+1)/2)} > 0.$$

Its pole remains a positive distance to the left of -1 uniformly for $s \in I$. This verifies the local hard/soft regularity used below.

25 The determinant ratio and logarithmic confinement

The primary analytic input is the background-weight ratio in Charlier–Gharakhloo [8, Proposition 9.1 and (9.1)–(9.4)], using its Section 5 transformations. With no bulk Fisher–Hartwig singularities, that ratio has an $O(N^{-1})$ logarithmic remainder. Their printed hypotheses impose superlogarithmic confinement; the potential (24.3) does not satisfy that condition. The following arguments adapt the ratio proof, rather than claim literal coverage by the printed theorem. They establish the remote-contour and differential-identity estimates at the precise places where stronger confinement was used.

25.1 Positive continuity at the off support cusp

Put $c = C^{-1}$ and $z_0 = [2^{3/2}(1 + \sqrt{2})]^{-1}$. Edgar's density formula [3, Proposition 23] gives

$$\varphi(c-) = \frac{4\sqrt{3}}{\pi(1+c)} \left[{}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; z_0\right) \right]^2 > 0. \quad (25.1)$$

The hypergeometric value is finite since $0 < z_0 < 1$. The local Frobenius branches at c have powers $0, 1/2, 1$ with no logarithms [3, Proposition 4]. Continue each upper and lower boundary branch used to define the jump density around c inside its own half-plane. The constant Frobenius coefficient is unchanged, and the square-root term vanishes in the limit. The density's right limit therefore equals (25.1). In particular,

$$\varphi(c \pm v) = \varphi(c) + O(\sqrt{v}) \quad (v \downarrow 0) \quad (25.2)$$

with convergent one-sided expansions. Thus φ , $\log \varphi$, f and the transformed background have local Hölder exponent $1/2$ at the cusp, with positive finite common limit.

At C , Edgar's Proposition 25 supplies a positive analytic factor times $\sqrt{C-x}$, yielding (22.2). At zero, Proposition 26 gives positive logarithmic growth. Consequently

$$h(x) = \exp(f((1+x)^{-1}))$$

extends positively to $x = 0$, has a positive finite cusp limit, and grows like a positive multiple of $\log x$ at infinity. In particular,

$$h(x) \geq h_{\min} > 0, \quad h(x) = O(1 + \log x) \quad (x \rightarrow \infty). \quad (25.3)$$

The off-support cusp fits the local Hölder hypothesis of [8]; no analyticity there and no surrogate density is needed.

25.2 Uniform remote jump estimates

Use the same symbol Q_s for the transformed effective potential, normalized by $Q_s(L_s) = 0$. From (23.1) or (24.5),

$$Q'_s(x) = \frac{s}{1+x} \sqrt{\frac{x-L_s}{x}}, \quad x > L_s. \quad (25.4)$$

Let $s_{\min} = \min I > 0$ and choose

$$X_0 > \max(1, 2 \max_{s \in I} L_s), \quad \eta_I = \frac{s_{\min}}{2\sqrt{2}}, \quad q_0 = \min_{s \in I} Q_s(X_0) > 0.$$

Then, uniformly for $s \in I$ and $x \geq X_0$,

$$Q'_s(x) \geq \frac{\eta_I}{x}, \quad Q_s(x) \geq q_0 + \eta_I \log(x/X_0). \quad (25.5)$$

On the compact remainder of the remote ray outside a fixed soft-edge disk, Q_s has a uniform strictly positive lower bound. This includes the cusp, whose distance from L_s is uniformly positive.

For the relative deformation take

$$W_\tau(x) = -b \log(1+x) + \log(1-\tau + \tau h(x)), \quad 0 \leq \tau \leq 1. \quad (25.6)$$

Since $h \geq h_{\min} > 0$, all these backgrounds are analytic in a common neighborhood of the compact union of supports and locally Hölder elsewhere. Uniformly in s, τ , for $x \geq X_0$,

$$\begin{aligned} x^b e^{W_\tau(x)} &\leq K_I(1 + \log x), & |\partial_\tau W_\tau(x)| &\leq K_I(1 + \log x), \\ |\partial_\tau [x^b e^{W_\tau(x)}]| &\leq K_I(1 + \log x). \end{aligned} \quad (25.7)$$

The global Szegő parametrices $P^{(\infty)}$ and their inverses are uniformly bounded on the remote ray, as are the needed parameter derivatives. The jump formula in [8, (5.15)] therefore gives

$$\|\Delta_N(x)\| \leq K_I(1 + \log x)e^{-NQ_s(x)}. \quad (25.8)$$

Its first τ derivative satisfies the same estimate up to a fixed additional power of $1 + \log x$.

For each fixed nonnegative integer ℓ and $p = 1, 2$, substitution $x = X_0 e^u$ in (25.5) gives the explicit bound

$$\int_{X_0}^{\infty} (1 + \log x)^\ell e^{-pNQ_s(x)} dx \leq e^{-pNq_0} X_0 \sum_{j=0}^{\ell} \binom{\ell}{j} (1 + \log X_0)^{\ell-j} \frac{j!}{(pN\eta_I - 1)^{j+1}}, \quad pN\eta_I > 1. \quad (25.9)$$

The supremum counterpart is exponentially small as well. Any fixed factor x^d replaces the denominator $pN\eta_I - 1$ by $pN\eta_I - d - 1$ and adds a fixed power of X_0 . These estimates give uniform exponential smallness in L^1 , L^2 and L^∞ for the remote jumps and their relevant derivatives.

25.3 Finite moments and the small norm problem

The transformed weight is $O(x^{-N(s+2)} \log x)$ at infinity. It need not have moments of every order at a fixed N . The orthogonal-polynomial problem of degree N uses only finitely many, and moments through degree $2N + k$ exist once $Ns_{\min} > k + 1$. This holds uniformly for every fixed k and all sufficiently large N . It also holds for the τ derivative. Positivity therefore gives the relevant orthogonal polynomials and nonzero determinants, and justifies the moment differentiation in the ratio identity.

On the endpoint circles the usual hard- and soft-edge parametrices have $O(N^{-1})$ matching errors. Together with (25.9),

$$\|\Delta_N\|_{L^2(\Sigma_R)} + \|\Delta_N\|_{L^\infty(\Sigma_R)} = O_I(N^{-1}).$$

The Cauchy projections on the ray/lens/circle contour have uniformly bounded L^2 operator norms: after affine normalization the contour shapes range over a compact regular family. The small-norm inverse therefore exists by a uniform Neumann series. If $\mu_R = I + C_-(\mu_R \Delta_N)$ is its boundary solution, then

$$R(z) = I + \frac{1}{2\pi i} \int_{\Sigma_R} \frac{\mu_R(\zeta) \Delta_N(\zeta)}{\zeta - z} d\zeta, \quad \|\mu_R \Delta_N\|_2 = O_I(N^{-1}). \quad (25.10)$$

It follows that

$$R(z) - I = O_I\left(\frac{1}{N\sqrt{\text{dist}(z, \Sigma_R)}}\right). \quad (25.11)$$

The L^1 estimates also yield the required expansion at infinity.

25.4 The differentiated tail without differentiating the cusp

The discarded part of the determinant differential identity requires a separate argument; jump smallness alone is not that argument. Let T be the first normalized Riemann–Hilbert transform in [8, Section 5]. On the right void set $v(z) = T(z)e_1$. The triangular T -jump fixes e_1 , so the column v is analytic through the remote real ray, including the cusp. Outside the support and lenses,

$$v = RP^{(\infty)}e_1.$$

For each remote real x , choose a fixed-radius circle centered at x that is separated from the support and endpoint disks. Estimate (25.11) has at most an integrable $|\text{Im } z|^{-1/2}$ majorant at its two real

crossings. Cauchy's integral formulas therefore give uniform bounds $v(x), v'(x) = O(1)$, also for arbitrarily large x . Bounded compact portions are handled by the same separated-circle argument.

Write the full transformed weight as $\omega_\tau(x) = x^b e^{-NV_s(x) + W_\tau(x)}$. The determinant-one property of the original orthogonal-polynomial matrix Y and reversal of its normalization give the exact cancellation

$$[Y^{-1}Y']_{21} \partial_\tau \omega_\tau = e^{-NQ_s(x)} (v_1 v_2' - v_2 v_1') \partial_\tau [x^b e^{W_\tau(x)}]. \quad (25.12)$$

The scalar equilibrium constant is included in Q_s . The two terms containing Ng' cancel in the Wronskian. No derivative of the Hölder weight occurs. By (25.7) and (25.9), the remote integral in the differential identity is $O(e^{-c_\tau N})$, uniformly in s and τ .

A contour enclosing the support must cross the real void ray. Choose that crossing in the analytic gap between the soft edge and cusp and locally remove the holomorphically continued, exponentially small triangular jump. On the continued contour, which is otherwise separated from Σ_R , (25.10) and Cauchy's formula give $R - I = O(N^{-1})$ and $R' = O(N^{-1})$. The compact-contour calculation in [8, Proposition 9.1], and the differential identities it uses from [9, (6.3)–(6.6)], now apply with the same $O(N^{-1})$ remainder. Integrating in τ proves the desired logarithmic ratio expansion. This completes the confinement adaptation: only the superlogarithmic tail input has been replaced, by explicit estimates.

25.5 Transforming the relative constant

The ratio calculation gives the equilibrium mean, a quadratic Dirichlet-energy term, and the hard-edge bias. We verify their normalization under (24.2). Parameterize the transformed support by $x = L_s(1 - \cos \vartheta)/2$, and denote by $\hat{f}_k^x$ the cosine coefficients of $f((1+x)^{-1})$ in that coordinate, with the same convention as (22.4). Put

$$q = \frac{1 - \sqrt{a}}{1 + \sqrt{a}} = \frac{1}{s + 1}.$$

The nonconstant cosine coefficients of $\log(1+x)$ are $-q^k/k$. Thus the cross term between $W_0 = -b \log(1+x)$ and the perturbation in the quadratic functional is $b \sum_{k \geq 1} q^k \hat{f}_k^x$. The disk automorphism induced by the Möbius map gives the Poisson identity and the conformal invariance identity

$$\hat{f}_0 = \hat{f}_0^x + 2 \sum_{k \geq 1} q^k \hat{f}_k^x, \quad \sum_{k \geq 1} k (\hat{f}_k^x)^2 = \sum_{k \geq 1} k \hat{f}_k^2. \quad (25.13)$$

These identities can also be proved first for harmonic polynomial or rational boundary data and extended by uniform analytic approximation. The transformed hard-edge bias is $b(\hat{f}_0^x - f(1))/2$; adding the cross term changes it to $b(\hat{f}_0 - f(1))/2$. Hence the final ratio is

$$\log \frac{D_N[t^{sN}(1-t)^b e^f]}{D_N[t^{sN}(1-t)^b]} = NF(s) + \frac{b}{2} (\hat{f}_0 - f(1)) + \frac{1}{2} \sum_{k \geq 1} k \hat{f}_k^2 + O_I(N^{-1}). \quad (25.14)$$

Taking $b = 1/2$ proves Theorem 22.1. The analytic neighborhoods, disks, lenses and void gaps may be chosen uniformly on I , as may all preceding bounds. This establishes compact-parameter uniformity as part of the adaptation; it is not inferred from a parameter-uniform statement in the printed theorem.

26 Barnes expansion and exact normalization tests

The recurrence of G converts (22.9) into the exact identity

$$S_N(r, b) = \frac{G(N+1)G(r+N+1)G(N+b+1)G(r+b+N+1)}{G(r+1)G(b+1)G(r+b+2N+1)}. \quad (26.1)$$

For fixed positive q and bounded d , the standard large-argument Barnes expansion gives

$$\begin{aligned} \log G(qN + d + 1) &= q^2 N^2 \left(\frac{\log N + \log q}{2} - \frac{3}{4} \right) + qdN(\log N + \log q - 1) \\ &\quad + \frac{qN}{2} \log(2\pi) + \left(\frac{d^2}{2} - \frac{1}{12} \right) (\log N + \log q) \\ &\quad + \frac{d}{2} \log(2\pi) + \zeta'(-1) + \frac{d^3/6 - d/12}{qN} + O(N^{-2}). \end{aligned} \quad (26.2)$$

The signed (q, d) pairs in (26.1) are

$$+(1, 0), +(s+1, 0), +(1, b), +(s+1, b), -(s, 0), -(s+2, b).$$

They cancel the unwanted $N^2 \log N$ and $N \log N$ terms. The coefficient of $\log N$ is $b^2/2 - 1/6$, equal to $-1/24$ when $b = 1/2$. The remaining constant factor is

$$\frac{(2\pi)^{b/2} e^{2\zeta'(-1)}}{G(b+1)} s^{1/12} (s+1)^{b^2/2-1/6} (s+2)^{-(b^2/2-1/12)}. \quad (26.3)$$

Combining this with (22.8) proves Corollary 22.2, including all factors in (22.10)–(22.12). At $b = 1/2$, the first term beyond the constant in the logarithm of the Selberg factor is

$$-\frac{1}{48N} \left(1 + \frac{1}{s+1} - \frac{1}{s+2} \right). \quad (26.4)$$

This is the Selberg contribution only, not the full first correction of the Apéry determinant.

There are useful exact tests of the ratio constant. For constant $f = c$, (25.14) reduces to Nc exactly. For $f(t) = \gamma \log t$, its left side is exactly $\log[S_N(sN + \gamma, b)/S_N(sN, b)]$. In this case

$$\hat{f}_0 = 2\gamma \log \frac{s+1}{s+2}, \quad \hat{f}_k = \gamma \frac{(-1)^{k+1}}{k(s+1)^k}, \quad k \geq 1.$$

The hard-edge bias and quadratic term are respectively

$$b\gamma \log \frac{s+1}{s+2}, \quad \frac{\gamma^2}{2} \log \frac{(s+1)^2}{s(s+2)}.$$

These are precisely the terms obtained by Taylor expansion of the exact Selberg quotient. They independently check the bias and the factor $1/2$ in the Fourier energy. A trace perturbation $f(t) = \lambda t$ gives limiting variance $(1-a)^2/16$ and mean bias $-b(1+s)/(2+s)^2$, in agreement with the exact Jacobi trace formulas.

27 Bounded phases and asymptotic inverse thresholds

Using $s = r/N$ exactly is essential. For a fixed $\rho > s_*$, write

$$r = \rho N - \delta_N, \quad |\delta_N| \leq D < \infty. \quad (27.1)$$

Taylor expansion of Corollary 22.2 gives

$$\begin{aligned} \log H_N(r) &= N^2 A(\rho) + N[B(\rho) - \delta_N A'(\rho)] - \frac{1}{24} \log N \\ &\quad + \log M(\rho) + \frac{\delta_N^2}{2} A''(\rho) - \delta_N B'(\rho) + O(N^{-1}). \end{aligned} \quad (27.2)$$

The error is uniform over bounded δ_N and compact admissible ρ ranges. In particular,

$$A'(s) = \log C + 2(s+1)\log(s+1) - s\log s - (s+2)\log(s+2), \quad (27.3)$$

$$A''(s) = \log \frac{(s+1)^2}{s(s+2)} > 0. \quad (27.4)$$

Since $A'(0+) = \log(C/4) > 0$ and $A(0+) = \log(C/4) > 0$, both A' and A are positive for $s > 0$. In fact, for $g(s) = 2A(s) - sA'(s)$,

$$g(0+) = 2\log(C/4) > 0, \quad g'(s) = \log \left[C \left(\frac{s+1}{s+2} \right)^2 \right] > 0.$$

For $r_N = \lfloor \rho N \rfloor$ and $k_N = r_{N+1} - r_N \geq 0$, expansion at the exact r_N/N gives the adjacent logarithmic increment $N[2A(\rho) + (k_N - \rho)A'(\rho)] + O(1)$. Its leading coefficient is $g(\rho) + k_N A'(\rho) > 0$, uniformly over these finitely many possible increments. Thus the full floor-shifted sequence is eventually strictly increasing, not merely each rational residue subsequence.

For $r = \lfloor \rho N \rfloor$, the phase is $\delta_N = \{\rho N\}$. For the older matrix-order convention $N = n + 1$ with $r = \lfloor \rho n \rfloor$, it is instead

$$\delta_N = \rho + \{\rho(N-1)\}. \quad (27.5)$$

The two conventions must not be interchanged. Dropping δ_N from (27.2) can change an exponential factor of size $e^{O(N)}$; it is not an innocuous change to the leading constant.

For a fixed phase δ , abbreviate

$$A = A(\rho), \quad B_\delta = B(\rho) - \delta A'(\rho), \quad C_\delta = \log M(\rho) + \frac{\delta^2}{2} A''(\rho) - \delta B'(\rho). \quad (27.6)$$

For a large target e^y , set $u = \sqrt{y/A}$ and

$$\mathcal{N}_\delta(y) = u - \frac{B_\delta}{2A} + \frac{B_\delta^2/(4A) + (\log u)/24 - C_\delta}{2Au}. \quad (27.7)$$

Substitution into $AN^2 + B_\delta N - (\log N)/24 + C_\delta$ leaves $O(\log u/u)$; division by its derivative $2Au + O(1)$ shows that the smooth inverse expansion has remainder

$$N_\delta(y) = \mathcal{N}_\delta(y) + O(\log u/u^2) = \mathcal{N}_\delta(y) + O(\log y/y). \quad (27.8)$$

This describes the inverse asymptotic model, with the forward $O(N^{-1})$ uncertainty included. A determinant indexed by matrix order is a discrete sequence; an integer inverse needs an additional statement.

Corollary 27.1 (Rational residue thresholds). *Let $\rho = p/q > s_*$ be rational in lowest terms, and take $r_N = \lfloor \rho N \rfloor$. For each $j \in \{0, \dots, q-1\}$ put $\delta_j = \{pj/q\}$. On the residue class $N \equiv j \pmod{q}$, the first sufficiently large index satisfying $H_N(r_N) \geq e^y$ has the form*

$$j + q \left\lceil \frac{\mathcal{N}_{\delta_j}(y) - j + \varepsilon_j(y)}{q} \right\rceil, \quad \varepsilon_j(y) = O(\log y/y). \quad (27.9)$$

The overall threshold is the minimum of these q candidates. For $r_N = \lfloor \rho(N-1) \rfloor$, replace the phase by $\delta_j = \rho + \{p(j-1)/q\}$.

Proof. On a fixed residue class the phase is constant. The leading difference of (27.2) between $N+q$ and N is $2AqN + O(1)$, so the sequence is eventually strictly increasing there. At its discrete threshold values, (27.8) locates the corresponding index within $O(\log y/y)$ of $\mathcal{N}_{\delta_j}(y)$. Between consecutive threshold values, the ceiling is fixed except when this approximation lies within that same error of a residue boundary. Choosing a bounded correction of that order inside the ceiling yields (27.9), including equality thresholds. There are finitely many residue classes; their earliest candidates give the overall threshold once y is large enough to exclude any fixed initial segment. $\square$

Remark 27.2. The sign and constant of ε_j are not determined here. Equation (27.9) is an asymptotic threshold formula, not a computable exact ceiling that is valid arbitrarily close to a threshold. Away from the stated shrinking neighborhoods of residue boundaries, the ceiling may safely use $\varepsilon_j = 0$. For irrational ρ , the nonperiodic floor phase remains part of the problem; no phase-free smooth inverse is claimed.

Context (editorial, batch 77P5). The passage from the smooth inverse (27.7) to the integer threshold (27.9) follows the pattern of the staircase theorem `p0:thm:staircase` of the repository's transseries volume [11]: its part (1) is the exact ceiling identity for a strictly increasing interpolation, its part (2) the separation condition behind the caveat above, and its part (4) the minimum over residue classes. The two-term smooth inverse is the first step of the perturbed inversion `p0:thm:perturbed-inversion` of the same volume. No Lambert- W block occurs here, so its `p0:thm:lambert-core` is not needed. No novelty is claimed for the inversion technique; what is specific to this Part is the forward expansion (27.2), with its phases, on which the inversion acts.

28 An explicit first correction

The full equivalent admits one more explicitly computable term. To state it unambiguously, hold the function f fixed and vary the left endpoint u near $a = a(s)$. Define

$$U(u) = \frac{1}{\pi} \int_0^\pi f\left(\frac{1+u}{2} + \frac{1-u}{2} \cos \theta\right) d\theta, \quad (28.1)$$

$$\widehat{f}_k[u] = \frac{1}{\pi} \int_0^\pi f\left(\frac{1+u}{2} + \frac{1-u}{2} \cos \theta\right) \cos(k\theta) d\theta, \quad S(u) = \sum_{k \geq 1} k \widehat{f}_k[u]^2. \quad (28.2)$$

Primes on U and S below are derivatives in u , not in s . All these derivatives exist with uniform bounds on the compact ranges used in this paper.

Theorem 28.1 (First correction). *Let*

$$c_{\text{rel}}(s) = \frac{\sqrt{a}(1-a)}{24s} \left[-4a(1-a)U'(a)^3 + 6aU'(a)^2 + 3U'(a) + 2aU''(a) \right], \quad (28.3)$$

$$c_{\text{full}}(s) = c_{\text{rel}}(s) - \frac{1}{48} \left(1 + \frac{1}{s+1} - \frac{1}{s+2} \right). \quad (28.4)$$

Uniformly for $s = r/N$ in every fixed compact $I \subset (s_*, \infty)$,

$$H_N(r) = C^{Nr+N(N-1)} S_N(r, \frac{1}{2}) e^{NF(s)} E(s) \left(1 + \frac{c_{\text{rel}}(s)}{N} + O_I(N^{-2}) \right), \quad (28.5)$$

$$H_N(r) = M(s) N^{-1/24} e^{N^2 A(s) + NB(s)} \left(1 + \frac{c_{\text{full}}(s)}{N} + O_I(N^{-2}) \right). \quad (28.6)$$

Equivalently, the logarithm of either ratio has the corresponding $c(s)/N + O_I(N^{-2})$ correction.

28.1 The finite order potential derivative lemma

For real ϵ near zero, let

$$V_{s,\epsilon}(t) = -s \log t - \epsilon f(t), \quad Z_N(s, \epsilon) = \mathcal{D}_N[e^{-NV_{s,\epsilon}(t)}(1-t)^b], \quad b = \frac{1}{2}.$$

Then $Z_N(s, 1/N)/Z_N(s, 0)$ is exactly the background ratio in Theorem 22.1. Let $[a_\epsilon, 1]$ and $\mu_{s,\epsilon}$ be the nearby equilibrium support and measure. For a function g , write $\langle g \rangle_{[a,1]}$ for its normalized arcsine average. The relevant coefficient functionals are

$$\Theta'_0(\epsilon) = \int f \, d\mu_{s,\epsilon}, \quad (28.7)$$

$$\Theta_1(\epsilon) = \log(2\pi) + b \log \frac{1-a_\epsilon}{4} - \frac{b}{2} \langle V_{s,\epsilon} \rangle_{[a_\epsilon,1]} + \frac{b}{2} V_{s,\epsilon}(1), \quad (28.8)$$

$$\Theta_2(\epsilon) = U_* - \frac{1}{24} \log(\pi \psi_{\text{soft}}(s, \epsilon)), \quad (28.9)$$

$$U_* = 2\zeta'(-1) + \frac{1}{4} \log(2\pi) - \log G(3/2).$$

Here ψ_{soft} is the analytic equilibrium-density factor at the soft edge after affine normalization to $[-1, 1]$, with the hard edge at -1 .

Lemma 28.2 (Uniform derivative expansion). *There exist $\epsilon_0 > 0$, N_0 and $K_I < \infty$, depending on I , such that for $s \in I$, real $|\epsilon| \leq \epsilon_0$ and $N \geq N_0$,*

$$\partial_\epsilon \log Z_N(s, \epsilon) = N^2 \Theta'_0(\epsilon) + N \Theta'_1(\epsilon) + \Theta'_2(\epsilon) + \mathcal{R}_N(s, \epsilon), \quad |\mathcal{R}_N| \leq K_I/N. \quad (28.10)$$

The displayed coefficient functionals are analytic in ϵ near zero, with uniformly bounded derivatives of every fixed required order.

Proof. We establish an estimate for the derivative identity itself; we do not differentiate an asymptotic remainder.

The local and global equilibrium family. The endpoint condition is

$$J(a, V) := \int_a^1 V'(t) \sqrt{\frac{1-t}{t-a}} \, dt = -2\pi. \quad (28.11)$$

At $\epsilon = 0$, $J_a = \pi s/(2a^{3/2}) \neq 0$. The analytic implicit-function theorem and the scalar Cauchy density formula therefore give analytic a_ϵ and local density data. The support remains uniformly above the cusp, and its analytic density factor remains positive for real small ϵ . The strict void inequality persists on compact remote intervals by continuity, while its positive $3/2$ soft-edge behavior persists immediately outside the moving edge.

In the Möbius coordinate the perturbed effective potential satisfies

$$Q_{s,\epsilon}(x) = s \log x - \epsilon \log \log x + O_I(1) \quad (x \rightarrow \infty),$$

uniformly after a harmless bounded change inside the logarithms. Choose a sufficiently large fixed cutoff. Since $s_{\min} > 0$, the $\epsilon \log \log x$ term is bounded by a chosen fraction of $s_{\min} \log x$, and the bounded remainder is absorbed by increasing the cutoff. It follows that $Q_{s,\epsilon}(x) \geq q + \eta_{I,1} \log(x/X)$ on the tail, for fixed $q, \eta_{I,1} > 0$. The estimates (25.9), with adjusted constants, thus remain valid uniformly in the real parameter. Only local spatial analyticity near the support is used: the remote cusp of the perturbed potential is confined to the exponentially small real-line jump.

One additional matching order. Choose uniform moving disks and lenses, or affinely normalize the support. The Airy/Bessel matching expansion on endpoint circles is

$$\Delta_N = \frac{\Delta_1(s, \epsilon)}{N} + O_I(N^{-2})$$

in both supremum and square-integrable contour norms, with analytic local coefficient data. The remote ray is exponentially small. One additional expansion of the small-norm equation gives

$$R = I + \frac{R_1(s, \epsilon)}{N} + O_I(N^{-2}), \quad \partial_z R = \frac{\partial_z R_1(s, \epsilon)}{N} + O_I(N^{-2}). \quad (28.12)$$

The coefficient R_1 is the Cauchy integral of Δ_1 ; its endpoint residue data are analytic in ϵ . A contour surrounding the support necessarily crosses the real void ray. Place its crossing inside the analytic gap between the soft edge and cusp. There the exponentially small triangular jump has a holomorphic continuation and may be removed locally. Cauchy estimates for the continued error matrix prove the derivative statement in (28.12) at the crossing as well. No continuation through the cusp is invoked.

Power counting and the tail of the exact identity. In physical coordinates the potential derivative multiplies the weight by Nf . Its remote contribution is controlled by the analytic-first-column Wronskian argument of (25.12). The factor N and the tail $f = O(\log \log x)$ are absorbed by the exponential bound; no spatial derivative of the cusp is taken. Substitution of (28.12) in the exact differential identity gives orders N^2 , N and 1 from the equilibrium, global-parametrix and first error-matrix terms. The residual $O(N^{-2})$ matrix error, multiplied by Nf on the compact contour, is $O(N^{-1})$.

Coefficient identification with moving endpoints. The local calculation is the potential-derivative computation in [8, Section 8, especially (8.28), (8.39), (8.43)–(8.44)]. Its residue identities are linear in a tangent potential variation and its induced density variation. Applied at each parameter value, they yield (28.7)–(28.9). The hard-edge order-one residue is proportional to $1 - 4b^2$ and vanishes for $b = 1/2$; the soft residue is $-(1/24)\partial_\epsilon \log \psi_{\text{soft}}$.

For clarity about covariance, use $t = (1 + a_\epsilon)/2 - q_\epsilon u$, $q_\epsilon = (1 - a_\epsilon)/2$. Heine's integral has affine factor $q_\epsilon^{N^2 + bN}$. Its derivative contributes only at orders N^2 and N , with no order-one term. The endpoint factor becomes $(1 + u)^b$, and the scaled soft density is precisely the ψ_{soft} in (28.9). Thus its total derivative includes the motion of the soft endpoint. One may differentiate in the original physical coordinate first and use this affine covariance only in the local contour formulas; there is no need to differentiate a moving remote cusp. This proves the lemma, including the uniform remainder and the claimed analyticity of the explicit coefficient data. $\square$

28.2 Eliminating the endpoint energy derivative

The correction ultimately requires only U' and U'' , through a general identity for a fixed analytic background.

Lemma 28.3 (Endpoint variation of the cosine energy). *For fixed real analytic f near $[a, 1]$, with U and S defined in (28.1)–(28.2),*

$$S'(a) = -\frac{1-a}{2}U'(a)^2. \quad (28.13)$$

Proof. Write $F_a(\theta) = f((1+a)/2 + (1-a)\cos\theta/2)$ and $c_k = \widehat{f}_k[a]$. Analyticity gives exponential decay of the coefficients and their endpoint derivatives, locally uniformly in a ; all subsequent series rearrangements and differentiations are therefore justified. Direct differentiation gives

$$\partial_a F_a(\theta) = -\frac{\tan(\theta/2)}{1-a} \partial_\theta F_a(\theta).$$

The apparent singularity at π is removable, since $\partial_\theta F_a = O(\pi - \theta)$ there. The finite trigonometric identity for $\sin(l\theta)/\sin\theta$ gives, for $l \geq 1$,

$$\frac{1}{\pi} \int_0^\pi \tan(\theta/2) \sin(l\theta) d\theta = (-1)^{l+1}.$$

Extend the integral by oddness to negative l , with value zero at $l = 0$. Product-to-sum then gives, for $k, l \geq 1$,

$$\frac{1}{\pi} \int_0^\pi \cos(k\theta) \tan(\theta/2) \sin(l\theta) d\theta = \begin{cases} 0, & l < k, \\ -1/2, & l = k, \\ (-1)^{k+l+1}, & l > k. \end{cases}$$

Consequently

$$U'(a) = \frac{2}{1-a} \sum_{l \geq 1} (-1)^{l+1} l c_l,$$

$$c'_k(a) = -\frac{k c_k}{1-a} + \frac{2}{1-a} \sum_{l > k} (-1)^{k+l+1} l c_l.$$

Multiply the second equality by $2k c_k$ and sum. Completing the square,

$$S'(a) = -\frac{2}{1-a} \left[\sum_{k \geq 1} (-1)^k k c_k \right]^2 = -\frac{1-a}{2} U'(a)^2.$$

□

This identity concerns general analytic f ; it is not a new sequence-specific asymptotic input.

28.3 Evaluation of the coefficient

The endpoint condition (28.11) becomes

$$s(a_\epsilon^{-1/2} - 1) + \epsilon I_f(a_\epsilon) = 2, \quad I_f(u) = \frac{1}{\pi} \int_u^1 f'(t) \sqrt{\frac{1-t}{t-u}} dt = (1-u)U'(u). \quad (28.14)$$

Consequently

$$D := a'_\epsilon(0) = \frac{2a^{3/2}(1-a)U'(a)}{s}. \quad (28.15)$$

Cosine diagonalization of the logarithmic kernel, the same quadratic energy as in (25.14), gives

$$\Theta''_0(\epsilon) = S(a_\epsilon), \quad \Theta'''_0(0) = DS'(a).$$

The endpoint condition also implies $\partial_a \langle V \rangle_{[a,1]} = -2/(1-a)$. It cancels the endpoint-motion terms in the first derivative of (28.8); hence

$$\Theta'_1(\epsilon) = \frac{b}{2} [U(a_\epsilon) - f(1)], \quad \Theta''_1(0) = \frac{b}{2} DU'(a). \quad (28.16)$$

For completeness, write the physical density as $\rho_V(t) = (2\pi)^{-1} \sqrt{(t-a)/(1-t)} h_V(t)$. The scalar Cauchy formula and integration by parts give

$$h_V(a) = -V'(a) + \frac{1}{\pi} \int_a^1 \frac{V'(t) - V'(a)}{t-a} \sqrt{\frac{1-t}{t-a}} dt = \frac{2J_a(a, V)}{\pi},$$

$$\pi \psi_{\text{soft}} = \frac{(1-a)J_a(a, V)}{2\pi}. \quad (28.17)$$

At the base model this equals $(s+1)(s+2)/s^2$. For $V_{s,\epsilon}$, $J_a/\pi = s/(2a^{3/2}) - \epsilon I'_f(a)$, so

$$\partial_\epsilon \log(\pi \psi_{\text{soft}})|_0 = -\frac{D}{1-a} - \frac{3D}{2a} - \frac{2a^{3/2}I'_f(a)}{s},$$

$$\Theta'_2(0) = \frac{D}{24(1-a)} + \frac{D}{16a} + \frac{a^{3/2}I'_f(a)}{12s}. \quad (28.18)$$

Integrate Lemma 28.2 over $0 \leq \epsilon \leq 1/N$. Its error integrates to $O_I(N^{-2})$. Taylor expansion of its explicit analytic coefficient functions, not of a remainder, gives

$$\begin{aligned} \log \frac{Z_N(s, 1/N)}{Z_N(s, 0)} &= N\Theta'_0(0) + \frac{1}{2}\Theta''_0(0) + \Theta'_1(0) \\ &\quad + \frac{1}{N} \left[\frac{1}{6}\Theta'''_0(0) + \frac{1}{2}\Theta''_1(0) + \Theta'_2(0) \right] + O_I(N^{-2}). \end{aligned} \quad (28.19)$$

The leading terms recover $NF + \log E$. The coefficient in brackets is

$$\begin{aligned} \frac{D}{6}S'(a) + \frac{D}{8}U'(a) + \frac{D}{24} \left(\frac{1}{1-a} + \frac{3}{2a} \right) \\ + \frac{a^{3/2}}{12s} [(1-a)U''(a) - U'(a)]. \end{aligned} \quad (28.20)$$

Inserting (28.15) first gives the equivalent energy form

$$c_{\text{rel}}(s) = \frac{\sqrt{a}(1-a)}{24s} [8aU'(a)S'(a) + 6aU'(a)^2 + 3U'(a) + 2aU''(a)]. \quad (28.21)$$

Lemma 28.3 then gives the displayed formula (28.3). Adding the exact Barnes term (26.4) gives (28.4). Exponentiation proves Theorem 28.1.

28.4 Two exact special cases

For $f(t) = \gamma \log t$, the coefficient reduces to

$$c_{\text{rel}}(s) = \frac{\gamma[-8\gamma^2 + 6\gamma s + 3s + 4]}{24s(s+1)(s+2)}. \quad (28.22)$$

Expanding the exact quotient $S_N(sN + \gamma, 1/2)/S_N(sN, 1/2)$ gives the same polynomial in γ , identically for $s > 0$. For a constant perturbation the correction is zero, as it must be.

There is an independent check for $f(t) = \lambda t$ using finite- N Jacobi/Selberg moments. If $T = \sum t_j$, its Schur averages satisfy

$$\mathbb{E} s_v(t) = s_v(1^N) \prod_{i \geq 1} \frac{(r + N - i + 1)_{v_i}}{(r + b + 2N - i + 1)_{v_i}}, \quad (28.23)$$

where $(z)_m$ is the rising factorial. This is the Schur specialization of the Selberg–Jack average, explicitly recorded in [10, equation (4.1)]. Partitions of sizes one, two and three give the first three trace cumulants. At $r = sN$, $b = 1/2$, their $1/N$ coefficients are respectively

$$\frac{s+1}{4(s+2)^3}, \quad \frac{s^2(s+1)}{2(s+2)^5}, \quad -\frac{2s^2(s+1)^2}{(s+2)^7}.$$

Formula (28.3) gives exactly

$$c_{\text{rel}}(s) = \frac{\lambda(s+1)}{4(s+2)^3} + \frac{\lambda^2 s^2(s+1)}{4(s+2)^5} - \frac{\lambda^3 s^2(s+1)^2}{3(s+2)^7}. \quad (28.24)$$

Thus its linear, quadratic and cubic responses agree separately with the exact mean, half-variance and one-sixth third cumulant contributions. The replay verifies these identities symbolically. They are checks of the coefficient algebra; the uniform derivative lemma supplies the analytic remainder.

28.5 The next inverse term

For the fixed phase δ in (27.1), define

$$c_\delta = c_{\text{full}}(\rho) - \frac{\delta^3}{6} A'''(\rho) + \frac{\delta^2}{2} B''(\rho) - \delta(\log M)'(\rho). \quad (28.25)$$

Then the sharpened logarithmic model is

$$y = AN^2 + B_\delta N + k \log N + C_\delta + c_\delta/N + O(N^{-2}), \quad k = -\frac{1}{24}.$$

With $u = \sqrt{y/A}$, set

$$d_0 = -\frac{B_\delta}{2A}, \quad d_1(u) = \frac{B_\delta^2/(4A) - k \log u - C_\delta}{2A}, \quad d_2 = -\frac{k d_0 + c_\delta}{2A}.$$

The refined inverse expansion is

$$N_\delta(y) = u + d_0 + \frac{d_1(u)}{u} + \frac{d_2}{u^2} + O\left(\frac{(\log u)^2}{u^3}\right). \quad (28.26)$$

Indeed $2Ad_0 + B_\delta = 0$ cancels the terms involving d_1/u at the next order, leaving $2Ad_2 + kd_0 + c_\delta = 0$. In Corollary 27.1, replace $\mathcal{N}_\delta(y)$ by the four displayed terms in (28.26) and sharpen the inside-ceiling error to $O((\log y)^2/y^{3/2})$. Its sign remains unspecified; the floor phase and threshold caveat remain.

28.6 Continuation to further fixed orders

The argument provides a continuation method on each fixed compact s -range above s_* . At any fixed order, Airy/Bessel matching supplies finitely many local jump coefficients; solving the singular-integral equation by its Neumann expansion supplies the corresponding finite list of error-matrix coefficients. Their dependence on the local potential data is analytic, and the remote estimates remain smaller than every algebraic power. The exact potential derivative identity can therefore be expanded to that order, with a uniform remainder, and integrated over $[0, 1/N]$ as above. This justifies continuation to any fixed number of inverse- N terms without differentiating a remainder. Only the first correction is evaluated explicitly in this report. A complete usable recursion for every determinant coefficient, together with its implementation, is not supplied.

29 Numerical evidence and reproducibility

The release contains a small standalone implementation of Edgar's upper-band hypergeometric density (shipped here as `code/03-shifted-density.py`). Its formulas and continuation convention are attributed both to Edgar [3] and to Part II's density program `code/02-szego-density_constant.py` (Appendix B); the replay reimplements only the upper-band density formulas needed for the proportional-shift constants. It does not include a copy of the repository or of its manuscripts. All quadrature nodes used here lie above the cusp.

For a specified rational s with integral $r = sN$, the numerical replay computes F , the cosine coefficients, the hard-edge bias and the Fourier energy. Independently it computes exact integer Apéry moments from

$$(k+1)^3 A_{k+1} = (34k^3 + 51k^2 + 27k + 5)A_k - k^3 A_{k-1}, \quad A_0 = 1, \quad A_1 = 5,$$

and performs high-precision elimination on the scaled positive Hankel matrix. The moment generation is exact; the subsequent elimination is floating point. The reported diagnostic is

$$\mathcal{E}_N(s) = \log H_N(sN) - [sN^2 + N(N-1)] \log C - \log S_N(sN, \tfrac{1}{2}) - NF(s) - \log E(s). \quad (29.1)$$

Theorems 22.1 and 28.1 predict $\mathcal{E}_N(s) = c_{\text{rel}}(s)/N + O(N^{-2})$.

At $s = 1$, using 512 midpoint cosine nodes, one obtains

$$F(1) \approx -0.9052804694622163, \quad E(1) \approx 1.3511218257963278.$$

Table 6 records the original numerical examples. For $s = 0.1$ and $s = 0.2$, 1024 nodes were used. The smaller-shift samples are closer to the collision and show longer preasymptotic behavior; no monotonic error claim is made.

Table 6: Logarithmic error after exact Selberg normalization

s	N	$\mathcal{E}_N(s)$	$N\mathcal{E}_N(s)$
1	20	0.00100562078	0.0201124156
1	40	0.000509359003	0.0203743601
1	80	0.000256302053	0.0205041642
0.2	20	0.000839079936	0.0167815987
0.2	40	0.000569348565	0.0227739426
0.2	80	0.000324954543	0.0259963634
0.1	20	0.00160846232	0.0321692463
0.1	40	0.00168695335	0.0674781341
0.1	80	0.000751788540	0.0601430832

Direct derivative quadrature of the proved coefficient gives

$$c_{\text{rel}}(1) \approx 0.0206331502033372131, \quad c_{\text{full}}(1) \approx -0.0036724053522183424.$$

The evaluations at 256 and 512 nodes agree within about 3.5×10^{-18} , a stability check rather than an interval bound. Subtracting $c_{\text{rel}}(1)/N$ gives the corrected residuals below. They are consistent with the uniform $O(N^{-2})$ remainder.

Table 7: Residual after the first correction at $s = 1$

N	$\mathcal{E}_N(1) - c_{\text{rel}}(1)/N$	$N^2[\mathcal{E}_N(1) - c_{\text{rel}}(1)/N]$
20	-0.0000260367287	-0.0104146915
40	-0.00000646975258	-0.0103516041
80	-0.00000161232494	-0.0103188796

The cosine transform uses double precision after high-precision density evaluation. Its decimal outputs therefore do not inherit the full precision of the determinant calculation. A small last cosine coefficient is a diagnostic, not a certified quadrature bound. The release also checks the critical value, equilibrium mass, Barnes cancellations and exact fixed-logarithm perturbation. Finite symbolic and numerical tests corroborate the normalization; they do not replace the analytic argument in Section 25.

30 Completion boundary and further questions

The full equivalent has an explicit constant; Theorem 28.1 adds a proved first correction and a compact-uniform relative $O(N^{-2})$ remainder. Several natural continuations remain outside that assertion.

1. **The cusp collision.** When $s \downarrow s_*$, the uniform analytic neighborhood used above collapses. A double-scaling analysis must account for an ordinary soft edge meeting a cusp of the fixed background density. The present proof gives neither its scaling law nor its transition function.
2. **The unshifted prefactor.** At $r = 0$ the equilibrium geometry and the endpoint/cusp singularities are different. The full prefactor of $D_n = H_{n+1}(0)$, including possible powers or oscillatory corrections, remains open in this report. Existing results of Parts I and II are not relabelled as new here.
3. **Further corrections.** The regular matching and potential-derivative construction gives the fixed-order continuation method in Section 28. Only c_{rel} and c_{full} are explicitly evaluated here. A complete implemented recursion for all higher coefficients remains to be developed. The residual table does not by itself identify or prove a second correction.
4. **Certified computation.** Interval evaluation of the density, Fourier tail and equilibrium integral would make the constants and finite-size comparisons numerically rigorous. The current release uses floating-point reproducibility, not interval enclosures.

Appendix A belongs to Part I (and is printed as it was); Appendices B and C belong to Part II. Part III has no appendix: its replay is described in Section 29 and in the README.

A Reproducing and auditing the package

The supplied files have separate roles. The article contains the proof. The file `data/determinants_0_200.txt` is not distributed with the archive; `build/apery_gmp200` rebuilds it. It contains exact integers; `code/apery_exact.py` is a standard-library Python implementation; `code/apery_gmp.cpp` is the faster C++17/GMP implementation; and `code/verify.py` performs the independent arithmetic checks. The plotting script additionally uses `mpmath` and `matplotlib`. The saved verification log records the actual checks performed.

From the package directory, the exact checks can be rerun with

```
python3 code/verify.py
```

A smaller exact table can be regenerated without external Python packages:

```
python3 code/apery_exact.py -n 40 -output data/recomputed.txt
```

For the full table, compile the supplied GMP implementation and run it with argument 200; the complete commands are in `README.md`. That program reports a failure rather than silently accepting a nonexact division or nonpositive pivot. `shift_experiments.py` similarly runs shifted examples and can use either implementation.

The symmetric elimination uses $O(n^3)$ integer arithmetic operations and $O(n^2)$ integer storage locations. These are *not* bit-complexity bounds: the integers grow with the determinant size. No claim of a faster asymptotic determinant algorithm is made. The PDF was compiled from this source and rendered for layout inspection. There is no Lean proof or external OEIS submission in the package.

B Reproducing the density evaluation

This appendix specifies the numerical continuation sheet; none of the main proofs depends on numerical continuation. Let $F(z) = {}_2F_1(1/3, 2/3; 1; z)$ and

$$q(x) = x^3 + 30x^2 - 24x + 1, \quad d(x) = x^2 - 34x + 1.$$

For $0 < x < c$, put

$$\lambda_{\pm}(x) = \frac{q(x) \pm (x^2 - 7x + 1)\sqrt{d(x)}}{2(x+1)^3}.$$

Edgar's hypergeometric representation gives

$$\varphi(x) = \frac{4\sqrt{3}}{\pi(x+1)} F(\lambda_{-}(x)) F(\lambda_{+}(x)). \quad (\text{B.1})$$

For $c < x < C$, set $s = 1/x$, $d_* = i\sqrt{-d(s)}$, and

$$M(s) = \frac{3 - 3s - d_*}{2(s+1)^2}, \quad z(s) = \frac{q(s) - (s^2 - 7s + 1)d_*}{2(s+1)^3}.$$

The continuation from the upper boundary of the Stieltjes transform is implemented as

$$\tilde{F}(s) = \begin{cases} F(z(s)), & s < (7 + 3\sqrt{5})/2, \\ F(z(s)) + i\sqrt{3}F(1 - z(s)), & s > (7 + 3\sqrt{5})/2. \end{cases}$$

Then the evaluated density is $-\pi^{-1} \operatorname{Im}(M(s)\tilde{F}(s)^2/x)$. The additional term accounts for crossing the hypergeometric cut; simply using the principal branch everywhere gives an incorrect density. These formulas specialize the representation in [3, Proposition 23].

Near the integration endpoints the implementation uses Part I's (2.2) and (2.3) at angular distances below 10^{-8} . This avoids subtracting nearly equal algebraic quantities or evaluating a zero-balanced hypergeometric function at a rounded argument equal to one. The cutoff and its unbounded numerical-error status are recorded explicitly in the source and output.

C Package contents and reproducibility

As delivered, the archive contained the manuscript's $\text{\LaTeX}$ source and its compiled PDF, the two vector figures (with PNG copies), four computational/plotting source files and a Makefile, and the exact and numerical data described in Section 19. No third-party paper, font file, precompiled executable, or proof-assistant certificate is included. The manuscript source and PDF are not shipped separately: their text is Part II of this report. The other files are shipped with the prefix `02-szego-`: programs and the Makefile in `code/`, data in `data/`, figures in `figures/`.

The delivered programs are kept byte-identical and still use their delivery names: `code/02-szego-verify_exact.py` writes `data/exact_checks.json`, `code/02-szego-make_figures.py` reads `data/norms.csv` and `data/constant.json` and writes `figures/ratio_limit.pdf` and `figures/monotone_envelope.pdf`, and `code/02-szego-Makefile` builds the unshipped `apery_hankel_refinement.tex`. Run them on a copy of the delivered layout, or pass explicit output paths, so that no shipped file is overwritten and no unprefixed file appears in the report. The report itself builds with

```
latexmk -pdf -interaction=nonstopmode apery_hankel_growth.tex
```

The finite exact test needs Python 3 (on a copy, as above):

```
python code/02-szego-verify_exact.py
```

The density calculation needs mpmath:

```
python code/02-szego-density_constant.py --dps 40 \
    --output <scratch>/constant.json
```

The high-precision matrix calculation needs a C++17 compiler and GMP:

```
g++ -O3 -std=c++17 code/02-szego-moment_norms.cpp \
    -lgmpxx -lgmp -o moment_norms
./moment_norms 200 4096 > <scratch>/norms.csv
```

The source accepts other cutoffs and precisions. For larger matrices the precision must grow to counter moment-matrix conditioning; passing the positivity checks alone is not a substitute for a precision comparison.

References

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